

A method for zone-level urban building energy modeling in data-scarce built environments

Orcun Koral Iseri^{a,*}, Ayca Duran^b, Ilkim Canlı^{a,c}, Cagla Meral Akgul^d, Sinan Kalkan^{e,f}, Ipek Gursel Dino^{a,g}

^a Middle East Technical University, Department of Architecture, Ankara, Türkiye

^b ETH Zurich – Federal Institute of Technology Zurich, Future Cities Laboratory, Switzerland

^c ODTU-GUNAM, Center for Solar Energy Research and Applications, Ankara, Türkiye

^d Department of Civil Engineering, Middle East Technical University, Ankara, Türkiye

^e METU-ROMER, METU Robotics and AI Technologies Application and Research Center, Ankara, Türkiye

^f Middle East Technical University, Department of Computer Engineering, Ankara, Türkiye

^g METU-CLIMATE, Climate Change and Sustainable Development Research and Application Center, Ankara, Türkiye

ARTICLE INFO

Keywords:

Urban building energy modeling
Building performance simulations
Data-scarce environments
Probabilistic data generation
UBEM resolution
Bottom-up UBEM

ABSTRACT

Urban Building Energy Modeling (UBEM) is critical for improving the resilience of cities to climate change, but most regions lack of data sets necessary for its development. A bottom-up approach is a viable method to initiate comprehensive UBEM frameworks. However, this process is often challenged by incomplete data, which can significantly affect the reliability and resolution of simulation results. Traditional deterministic approaches commonly used in UBEM fail to capture the diversity of the building stock. Thus, probabilistic methods are increasingly used, which require a careful examination of the types and patterns of missing data. This paper fills a critical gap in the literature by presenting a probabilistic approach to data generation for data-scarce environments to build high-resolution bottom-up urban-scale models while preserving building stock heterogeneity and statistical consistency. Our methodology includes advanced data imputation and generation techniques based on density estimations. This approach is illustrated with a case study in the Bahçelievler neighborhood in Ankara, Turkey. We have developed four different UBEM versions with varying degrees of data granularity to demonstrate the effectiveness of our methods. The proposed models incorporate comprehensive data on construction and occupant-related parameters, enhancing the resolution of energy simulations for buildings. This research provides a robust framework for the development of UBEM in regions lacking comprehensive datasets, ultimately supporting informed policy making and improved urban energy management.

1. Introduction

Urban regions are responsible for more than two thirds of global energy demand and more than 70% of global CO₂ emissions [1]. Buildings in developed countries in particular are responsible for around 40% of this energy use. The UN anticipates a rise in urban population to 5 billion people by 2030, which will significantly increase the energy demand of the building stock [2]. Furthermore, the IEA warns that energy demand is expected to increase by 50% by 2050 if energy efficiency measures are ignored [3]. To address these challenges, a comprehensive understanding of current building conditions is essential before implementing mitigation and adaptation strategies [4,5]. Concurrently,

studying the energy performance of the building stock can provide valuable insights to stakeholders by helping them formulate energy strategies and promote sustainable practices [6]. In this regard, UBEM can be the optimal solution for the examination process as it reveals the complex interdependencies between the urban building stock and its surroundings.

UBEM uses an engineering approach that employs extensive data sets and climate-adapted heat flow models to analyze the building stock and formulate data-driven energy policy measures at different levels. UBEM can easily capture the dynamic interplay between buildings and their urban microclimate by addressing several factors including solar reflectance, radiation or the effects of urban heat islands (UHI) [7]. The

* Corresponding author.

E-mail addresses: orcunkoraliseri@gmail.com (O. Koral Iseri), duan@arch.ethz.ch (A. Duran), ilkim.canli@metu.edu.tr (I. Canlı), cmerral@metu.edu.tr (C. Meral Akgul), skalkan@metu.edu.tr (S. Kalkan), ipekg@metu.edu.tr (I. Gursel Dino).

<https://doi.org/10.1016/j.enbuild.2025.115620>

Received 2 January 2025; Received in revised form 12 March 2025; Accepted 15 March 2025

Available online 20 March 2025

0378-7788/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

modeling leads to energy-efficient urban design and planning while conducting spatial and temporal analyses of energy supply and demand in buildings, e.g. distributed energy demand and the impact of renewable energy [8]. UBEM can forecast the impact of new technologies on energy demand even in the absence of building energy records that can be used for calibration [9].

1.1. Different modeling approaches in UBEM

Top-down and bottom-up models are the two main modelling categories of UBEM. Top-down models require aggregated historical data to estimate building performance based on correlations between building performance and various socio-economic, technical and physical factors [10]. However, the need for extensive and consistent data and the limited ability to describe specific spatial or temporal aspects pose a major challenge [11]. Therefore, the use of top-down models is limited to cities that have comprehensive building data [7]. Bottom-up models bottom-up models are disaggregated, physics-based simulation models that can provide extended information on the different performance objectives about building(s) e.g. the impact of energy efficiency improvements on CO₂ emissions and the identification of cost-effective carbon reduction strategies [12]. This is because they are based on complete energy flow models that are influenced by many different external factors. In contrast to top-down models, bottom-up models can provide information on different time scales when analysing energy supply and demand scenarios and forecasting future energy demand, while top-down models can only provide specific time scales based on available historical data [13].

Bottom-up modelling requires a comprehensive data set for performance modelling, including the physical characteristics of the building (geometry, thermal properties of materials), energy systems (HVAC, lighting, appliances), occupancy patterns, and internal loads. These data types can be grouped under two categories: geometric and semantic data. On the other hand, these datasets often come from independent sources, i.e. national databases, Energy Performance Certificates (EPCs) and metering data. However, they are usually either unstructured or incomplete for UBEM. Furthermore, when multiple data sources are to be combined, problems with data inconsistency or redundancy often occur [14]. One of the biggest challenges for UBEMs is the acquisition and structuring of the data format, i.e. missing data can lead to problems in data analysis, i.e. misinterpretation and biased results [15]. Therefore, the following sections discuss the different data inputs for UBEM and the existing methods for handling incomplete or missing data.

1.2. Data inputs for UBEM process

In the complicated process of UBEM, the accuracy and robustness of the results depend on the detailed representation of both semantic and geometric data inputs, both of which play a crucial role in for reliability of the simulation. Since improving the thermal performance of buildings is one of the most important measures in energy retrofit, representing the accuracy and heterogeneity of building performance in UBEM is essential [14]. This accuracy is achieved through a comprehensive semantic, geometric data representation. Geometric data provides the spatial context of buildings, which forms the basis for the modeling process as all other parameters are mapped to these geometric definitions. Semantic data includes construction attributes, conditioning systems, and their controls, and occupant-related parameters. In addition, weather data is a critical component that influences the thermal behavior and energy demand of the building. Together, these data types form the basis of UBEM and enable a comprehensive understanding of energy dynamics in urban buildings, especially in data-scarce environments [16].

1.2.1. Semantic data characteristics and alternative modeling approaches

Semantic data is crucial for the accurate representation of building

performance in UBEM [17]. This includes building construction parameters such as thermal properties (U-values) and insulation properties, which are crucial for evaluating energy use and efficiency. HVAC systems and their controls, such as setpoint/setback or ventilation strategies, also play an important role, but obtaining comprehensive data for these systems is often a challenge [18–20]. In cases where real-world data is scarce, statistical approaches can be applied to generate a generalized representation of the semantic data. However, given the diversity of building types in urban contexts, relying solely on building standards is not sufficient to implement the diversity of semantic data required for UBEM. Occupant behavior, especially in the control of systems such as ventilation and lighting, has a significant impact on energy consumption. Therefore, representing the occupants of a building at the city level is essential for realistic energy modeling [21,22]. These interactions between occupants can lead to variations in energy demand of up to 30 %, a significant factor that exceeds the influence of other variables such as infiltration [23]. While advanced techniques such as sensor networks provide promising results in detecting occupants in real time, the application of these methods at the city level leads to complex data collection issues due to the increased population size [24,25].

1.2.2. Geometric data and its role in spatial representation

Geometric information provides a comprehensive volumetric representation of building structures, i.e. building footprints, number of floors, window-to-wall ratios and terrain features. These are important features for energy interactions that depend on the physical dimensions. Therefore, the precision of the 3D geometric definitions is crucial as it has a direct influence on the definition of the thermal zones. The accurate specification of building volumes as well as the consideration of neighboring elements such as surface proximity, subsurface allocations, and thermal bridge calculations depends on a robust 3D geometric model. The representation of the building form varies in detail. For example, oversimplified models, e.g. shoebox approaches, often lead to significant discrepancies in energy efficiency due to the complexity of the interior [26]. In contrast, detailed modeling in extensive UBEM scenarios leads to increased computational effort, as the complexity of heat transfer and shading calculations increases proportionally to the number of surfaces of the city model.

Thermal zoning plays an important role in geometric data modeling, as it is closely linked to the details and complexity of building forms. This zoning facilitates a nuanced representation of occupant characteristics, behaviors and internal load distributions and improves the granularity of building performance simulations. Different zoning strategies each contribute differently to the level of detail and precision of the simulation [27,28]. These approaches range from single-zone models to more complex approaches such as (i) one zone per floor, (ii) perimeter/core (division of a floor into a core and a number of perimeter zones for each orientation, as also recommended in ASHRAE 90.1 Appendix G), (iii) one zone per room, the (iv) prototype method (i.e. prototype geometries and scales are used for the same footprint, orientation and aspect ratio as the target buildings). As with the selection of building form, a higher resolution in thermal zoning increases the representational capability and accuracy of the simulations [29]. However, this higher level of detail poses a challenge when managing a large number of buildings, as comprehensive floor plans are often not available, especially for large-scale UBEM implementations. This creates a delicate balance between the precision of the model and practical aspects such as the size of the model, the complexity and the computational intensity of the simulations, especially when applying complicated zoning methods such as the one-room-one-zone approach in urban scale projects [30]. Consequently, a critical equilibrium is required that reconciles an acceptable degree of simplification with the retention of essential performance-related data.

1.2.3. Strategies for data collection for UBEM

Data collection in UBEM includes the collection and processing of semantic, geometric and weather-related data, each of which presents unique challenges and methods. Semantic data collection for existing buildings is a challenge because the information is often not publicly available throughout a building’s lifecycle. To solve this problem, in-situ measurements such as heat flux meters and infrared thermography techniques can be used, albeit with practical limitations due to their extensive testing procedures and technical requirements [31]. For geometric data, on the other hand, there are more possibilities. 3D representations in CityJSON or CityGML formats are very powerful for urban microclimate analysis [32], but obtaining these data at the urban scale is challenging. Methods include the use of planning data, field measurements and resources such as OpenStreetMap for building footprints and approaches such as nDSM for building heights. However, details such as WWRs are often based on building standards or expert opinion in the absence of comprehensive databases [13,33,34]. Access to such data is limited, especially in developing countries, and the integration and processing of this data requires a considerable amount of time and effort. Finally, weather data are crucial for simulating environmental interactions and obtaining them is the easiest as there are many studies in the literature. For example, Typical Meteorological Year (TMY) data, real weather data, and future weather forecasts are the primary sources based on databases such as the U.S. DOE or software such as Meteonom [35]. Real weather data, which is essential for more localized studies, can be obtained from local weather stations or commercial websites [36]. For future weather data considering urban heat island effects and climate change, tools such as the Urban Weather Generator and the Morphing method are used, which rely on real weather data and models such as GCM or RCM [35]. Each type of data in UBEM, from semantic to geometric to weather data, requires careful consideration of source reliability, accessibility, and the balance between detail and computational feasibility, reflecting the complex nature of energy modeling in urban environments.

1.3. Existing approaches for data generation in UBEM

The lack of comprehensive and reliable data sets for feeding bottom-up UBEMs is a significant obstacle to the creation of valid energy models [37]. In parallel, the uncertainties associated with data availability (i.e. accessibility) and quality have necessitated the formulation of methods based on various assumptions, especially in multi-building contexts [38]. These methods can be broadly divided into two categories: deterministic and probabilistic approaches, as shown in Table 1.

First, the deterministic approaches use a single value as input without considering noise or uncertainties caused by a lack of information [39]. A common strategy is archetype characterization, i.e., the reductive modeling approach. The method clusters the number of buildings into archetypes and thus reduces the number of buildings to be processed [36]. Buildings that have similar characteristics (e.g. age, function, floor area, shape) are assigned to similar clusters. Thanks to this approach, a single number of building datasets can be created for each archetype (i.e. each cluster) [39–41]. The representative dataset can either be derived from the existing literature or extracted from the existing buildings in the neighborhood. Filogamo et al is suggested that

the segmentation of buildings with appropriate characterization is significantly helpful in analyzing the energy demand of heating, lighting and appliances to represent large building stocks [42]. Davila et al developed a UBEM for Boston based on GIS data by dividing buildings into 52 archetypes based on their function and date of construction [9]. Monteiro et al categorized the buildings into archetypes based on their construction, size, roof type, and physical urban context and assigned the mean values of each archetype to the missing properties [43]. Pasichnyi et al created archetypes based on the total number of buildings, heated area, absolute and specific energy, and peak power determined from measured data [44]. Similarly, Iseri and Dino used machine-learning clustering algorithms to characterize archetypes based on the thermal and physical properties and performance objectives of buildings in the selected neighborhood [45]. The main advantage of archetype characterization is that it eliminates the need for data collection at the individual building level and simplifies data collection [39]. However, clustering should be performed with sufficient data sets. Otherwise, clustering based on a limited number of inputs may ignore the complex nature of the input variables and lead to incorrect results [37]. Although archetypes are useful for gaining a general insight into the building stock, their practical utility is limited when detailed building-level modeling information is required. If the archetypes are characterized deterministically, the low data resolution can lead to an oversimplified and biased characterization of the archetypes [44].

The probabilistic approach, on the other hand, aims to preserve the heterogeneity of building data by managing noise and parameter uncertainties caused by the unavailability of data. This method increases the robustness in estimating the energy demand of existing buildings and in determining appropriate retrofit strategies [46]. Cerezo et al compared the deterministic and probabilistic approaches to characterize archetypes and concluded that the probabilistic approach has a lower standard deviation for the error between the calculated and actual energy use intensity [47]. Similarly, Sokol et al [39] applied two approaches to characterize archetypes and showed that the probabilistic approach leads to more accurate results for energy use intensity. Wang et al harmonized the data from different sources and proposed distributions for the missing data according to the created archetypes based on the type of use and year of construction [37]. Lim and Zhai found that deterministic archetype characterization of archetypes is not able to represent different design alternatives and proposed a combination of two approaches [39]. Finally, there were a limited number of studies that used non-archetype approaches with probabilistic parameters. Nägeli et al created synthetic building stocks using distributions for the envelope parameters [48]. Mastrucci et al presented simplified energy models for the building stock by considering the uncertainty of the parameters with different types of probability distributions [49]. Kristensen et al defined different distributions for 32 unknown parameters of the geometric representation [46].

Lack of data can significantly affect the quality of conclusions and limit the ability to draw accurate conclusions. Although UBEM has been successfully applied in these studies, the problem of missing data is an ongoing challenge [50]. This problem hinders the creation of high-resolution, realistic building models and leads to inaccuracies in the simulation results. A notable solution to this problem lies in the use of data imputation and generation methods. When imputing data, missing values are estimated and completed using statistical or machine learning methods [33]. These methods have been applied in UBEM for datasets ranging from electricity demand to district heating and building dimensions [51–53]. However, the effectiveness of these imputation methods depends on understanding the patterns of missing values in the data [54]. As each pattern of missing data may require a different imputation technique, a thorough analysis of both the missing and existing data is crucial to ensure the selection of the most appropriate and accurate imputation methods.

Table 1
Data generation approaches in the literature.

Data generation approach	Approach
Deterministic [36,39]	Archetype characterization from literature Archetype characterization from building data
Probabilistic [37]	Archetype characterization with probabilistic parameters Non-archetype approaches with probabilistic parameters

1.4. Aims and scope of the study

The incompleteness of the data in the bottom-up development of UBEM significantly affects the reliability of the simulation results. To reduce the degree of uncertainty in these results, it is therefore essential to solve the problem of missing data. While deterministic methods are widely used in UBEM, they often fail to adequately represent the diversity of the building stock, so probabilistic approaches need to be explored. However, these approaches require careful examination of the different types and patterns of missing data. A notable gap in the existing literature is the lack of methods that systematically generate building datasets considering the different mechanisms of missing data to ensure the heterogeneity of building stocks and their statistical integrity. Therefore, in this study we propose a methodology for the bottom-up development of UBEM in scenarios where complete and high-quality building datasets are missing. The focus is on the generation or imputation of data that includes the degree of missing data and density estimation. The proposed approach supports the development of zone-level energy models where semantic data related to occupants, zone-level electricity demand and occupant preferences (i.e. set points) can be considered.

Our method is illustrated with a case study in a neighborhood in Ankara, Turkey. It includes the collection of data from various sources, the application of methods for imputation and generation of data based on density estimation, and the development of a high-resolution UBEM. Four different UBEM versions with increasing data granularity are developed. The baseline as a first model is established upon standard data definitions of materials, occupants and systems. The second and third models are developed using the data generated by the proposed approach with the occupant-related and building-related (i.e. construction, HVAC) information. The fourth model is the most comprehensive model that includes both occupancy-related and building-related information. Simulations are performed for each model version and then performance objectives are calculated and compared.

2. Empirical context and data collection

2.1. Description of case study

The application of the proposed method takes place in the Bahçelievler district of Ankara, Turkey. The selected area covers approximately 0.57 km², as shown in Fig. 1. Ankara has warm and dry summers and cold and snowy winters. The climate of the city belongs to the CSB Koppen climate zone and the ASHRAE 4B climate zone, with HDD18.3 °C = 3567 °C-day and CDD23.3 °C = 119 °C-day. The selection of this area results from the predominant residential development and the diversity of thermal characteristics of the buildings.

The construction date of the buildings varies from 1950 to 2022 and the simulation process includes detailed modelling and analysis of the residential buildings. There are 593 residential buildings of the 642 buildings in the area while the remaining commercial and public

buildings. These buildings consist of an average of 3.9 floors and comprise 11.8 residential units each. Brick masonry predominates in the older buildings, while concrete construction has been more common since the 1970s. Each building consists of several residential units with an average living space of 117.7 m², most of which are occupied by one family. The development of UBEM occurs at zone level, with each entire residential unit treated as a thermal zone without any room-partitioning.

2.2. Local data sources for data imputation

Table 2 shows the diverse sources for the collection of geometric and semantic data. We obtained master plans and other building details, such as building layout and entrance directions, from the Ankara Metropolitan Municipality GIS databases. The Ministry of Interior's Address Inquiry System (AIS) (Tr. Adres Kayıt Sistemi) provided us with information including address records, number of floors and total number of residential units for each building. In the selected region, energy auditors have recorded Energy Performance Certificates (Tr. EKB, Enerji Kimlik Belgesi) for most of the buildings, which we retrieved from the Ministry of Environment, Urbanization and Climate Change [55]. These EPCs contain information on construction materials and energy demand measurements. However, since EPCs are only issued for property sales or rentals, they are only available for 30% of buildings. Finally, we used census data from the Turkish Statistical Institute (TUIK) to collect occupant-related information [56].

Three important geometric data elements form the basis for the development of a 3D model in UBEM, and for this study each of them came from different sources. The Master Plan of the Municipality (MPM) provides $B_{Footprint}$ and N_{Floor} , while the Address Inquiry System (AIS) provides N_{Zone} . Semantic data, on the other hand, refers to the building level and includes elements such as the WWR, construction date and building or HVAC systems, or refers to the zone level with occupant-related information. The EPCs contain the WWR. The municipality's GIS records contain the construction date and building type (masonry or concrete), with no missing data. In contrast, the rate of missing data is 70 for building envelope materials or energy systems, which primarily come from the EPCs.

In cases where household data is completely missing at the zone level, TUIK TurkStat's Family Statistics helped to estimate the number of residents per zone through parametric density estimation [57]. Information on building-level metrics such as SHGC, COP_{Boiler} and I_{rate} is completely missing, as well as data on zone-level parameters such as $Q_{Equipment}$, $Q_{Lighting}$ and T_H are not available for all buildings, with a 100 rate of missing data.

3. Methods for data imputation and generation

This study presents a method for bottom-up development of UBEM in data-scarce environments as shown in Fig. 2. Data imputation and generation are the main approaches for developing complete data sets



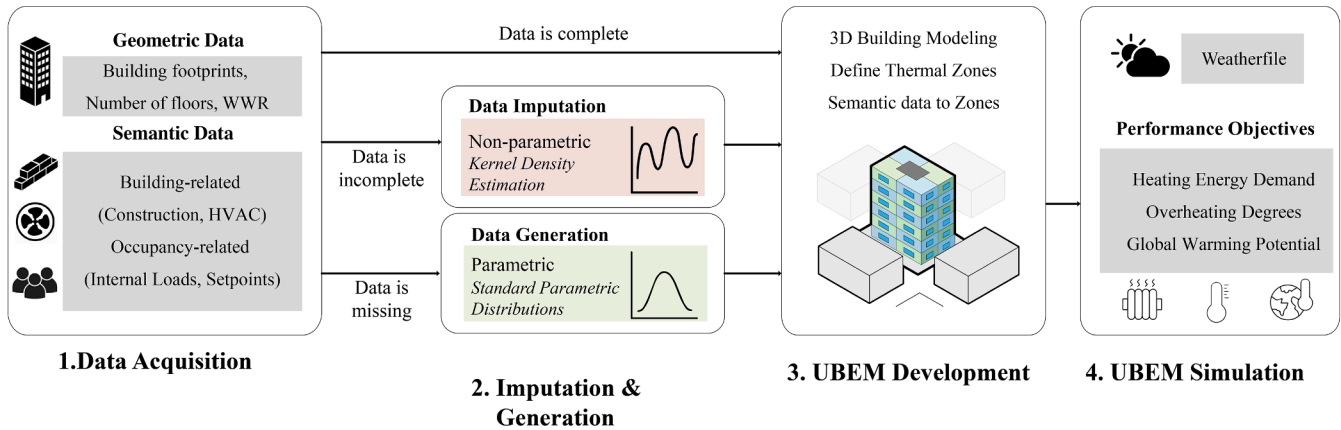
Fig. 1. (a) construction date, (b) EPC (EKB) existence, (c) building function and (d) construction material.

Table 2
UBEM data inputs.

	Abbrev.	Name	Unit	Associated	Source	Missing	Imputation/generation method
Geometric	B _{Footprint}	Building footprint		Building	MMP	0 %	—
	N _{Floor}	Number of floors		Building	MMP	0 %	—
	N _{Zone}	Number of zones		Building	AIS	0 %	—
Semantic	C _{Date}	Construction date	Year	Building	MGIS	0 %	—
	C _{Type}	Construction type	—	Building	MGIS	0 %	—
	WWR	Window-wall ratio	%	Building	EPCs	70 %	NPDE
	U _{Wall}	U-value, wall	W/m ² K	Building	EPCs	70 %	NPDE
	U _{Roof}	U-value, roof	W/m ² K	Building	EPCs	70 %	NPDE
	U _{Ground}	U-value, ground floor	W/m ² K	Building	EPCs	70 %	NPDE
	U _{Window}	U-value, window	W/m ² K	Building	EPCs	70 %	NPDE
	SHGC	Solar Heat Gain Coefficient	%	Building	—	100 %	PDE
	COP _{Boiler}	Boiler efficiency	%	Building	—	100 %	PDE
	I _{rate}	Infiltration rate	m ³ /s per m ²	Building	—	100 %	PDE
	Q _{People}	Occupant density	ppl/m ²	Zone	TUIK	100 %*	NPDE
	Q _{Equipment}	Equipment load density	W/m ²	Zone	—	100 %	PDE
	Q _{Lighting}	Lighting load density	W/m ²	Zone	—	100 %	PDE
	T _H	Heating setpoint	°C	Zone	—	100 %	PDE

MMP: Municipality master plan, MGIS: Municipality GIS records, AIS: Address Inquiry System, EPCs: Energy performance certificates.

* This value is only available on the city scale, not neighborhood scale.

**Fig. 2.** Proposed workflow for UBEM development.

and integrating them into energy models at the urban-scale.

A statistical method based on density estimation is used to impute data [33]. Density estimation produces an estimate of an unobserved underlying probability density function, with the observed data serving as the primary source of information [58]. Once the study estimates a density distribution, it generates new sample points based on this distribution to fill the missing data sets. Two approaches to density estimation are used in the study: non-parametric (NPDE) and parametric density estimation (PDE).

In PDE, the density is assumed to belong to a known parametric family of distributions, i.e. normal or uniform, with unknown parameters. The challenge is then to determine the most appropriate parameters that define this distribution. The advantages of PDE include its simplicity and the minimal number of data points required to determine the distribution parameters. However, practical data observations rarely match perfectly with standard parametric families. This mismatch can lead to poor predictive performance due to the strict distributional assumptions of PDE. More often, the data sets exhibit uneven, asymmetric or discontinuous distributions. In contrast, NPDE constructs the density function using samples from the distribution in question. This method is suitable for data sets with varying degrees of missingness, from 0% to 100%. Therefore, NPDE is often considered a more universal approach.

3.1. Data imputation with non-parametric density estimation (NPDE)

First, we selected the data sets with $0 < \text{missingness} < 100\%$ (a and

Q_{people}) and plotted the frequency distributions (Fig. 3). For the data sets that do not follow any of the known parametric distributions, we use kernel density estimation (KDE). This is basically an NPDE method that estimates a density distribution by averaging over homogeneous kernel functions centered at each sample point (1, 2). The kernel-based estimator with kernel K is defined by:

$$\hat{f}(x) = \frac{1}{Nh} \sum_{i=1}^N K\left(\frac{x - x_i}{h}\right) \quad (5)$$

The function $\hat{f}(x)$ symbolizes the estimated density function, which is derived from the sample data by kernel density estimation and serves as an approximation to the true but unknown density function $f(x)$. This calculation involves summing the kernel functions K centered at each data point x_i and then normalizing by the product of the number of points N and the bandwidth h . The bandwidth determines the smoothness of the density estimate, and the normalization ensures that $\hat{f}(x)$ is a proper density function that integrates to one.

The study chooses a Gaussian kernel function and uses Silverman's rule of thumb for bandwidth selection to capture the variance [59]. The method weights each data point in the distributions equally and assigns equal importance to the different building features. Gaussian distributions, unlike Box, Tri- or Tri-weight kernels, never reach a probability of zero, which leads to the creation of data points outside the observation

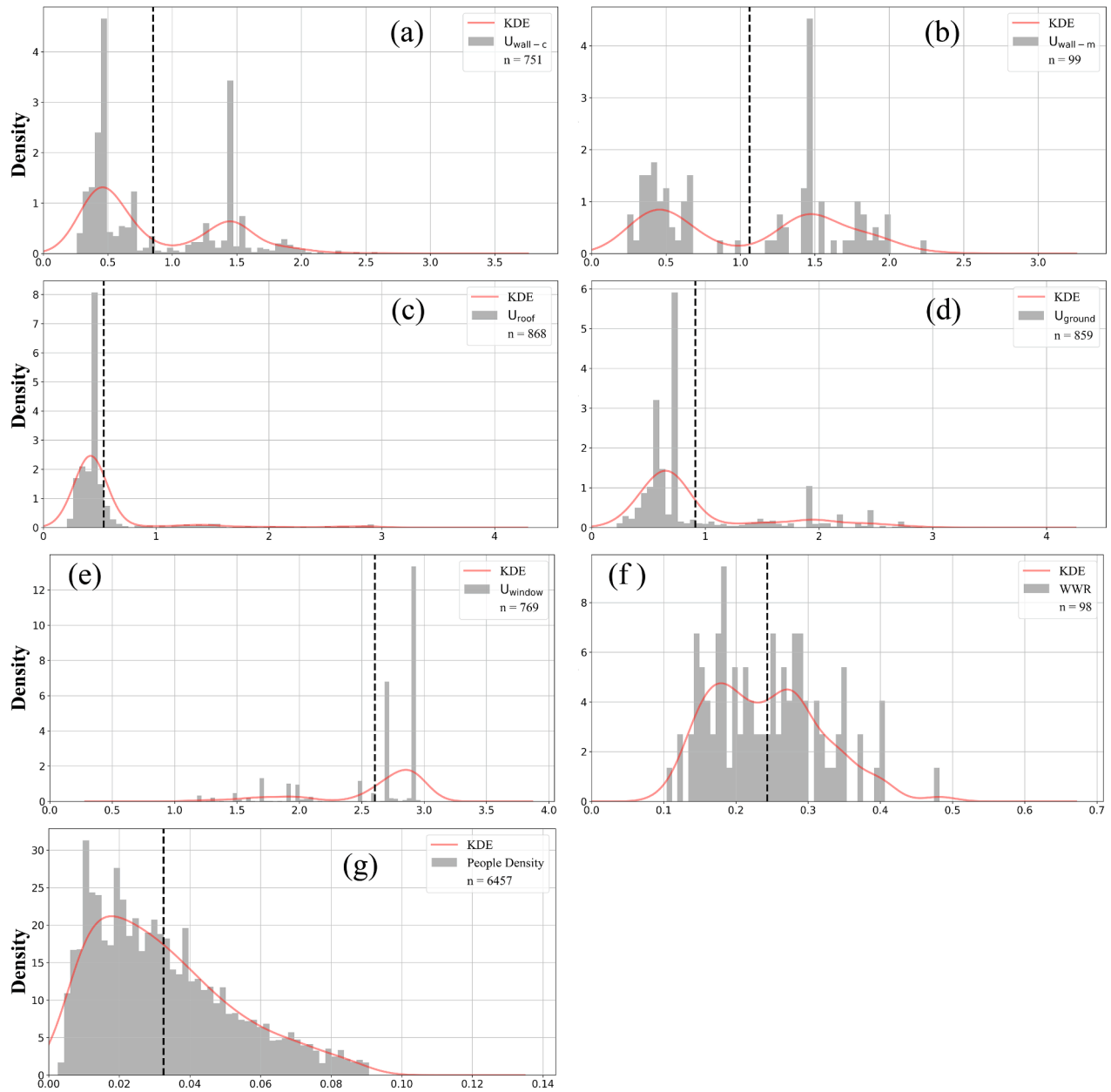


Fig. 3. KDE for (a) $U_{\text{wall-concrete}}$ (b) $U_{\text{wall-masonry}}$ (c) U_{roof} (d) U_{ground} (e) U_{window} (f) WWR (g) People density. Vertical dashed lines represent mean values.

range. The method regenerates data points outside the original data set until they lie within the specified minimum and maximum values as shown in Fig. 3.

The dataset sizes for each parameter differ from the total number of 593 residential buildings in the case study. The data sizes shown in Fig. 3 are larger for the subplots, except for WWR. The additional data points are incorporated for a more comprehensive distribution estimation. This expansion was necessary because the study focuses on the Bahçelievler neighborhood, however the available data from the neighbourhood is insufficient. Therefore, additional data was incorporated from the adjacent neighborhoods of Yukari Bahçelievler and Emek which share similar building construction styles and typologies.

The analysis of the data sets shows that both masonry ($n = 784$) and concrete ($n = 99$) play a decisive role in the distribution of the U-values of the building envelope. The predominant use of masonry until the 1970s and the subsequent transition to the use of concrete leads to different material sets for the two building types. Therefore, separate core density estimates are required for their envelope U-values. However, for the window-to-wall ratio (WWR) and occupant density, the study finds no significant difference, resulting in a uniform kernel density estimate for both masonry and concrete buildings.

Population density data (ppl/m^2) are the number of people per square meter in the thermal zone, and they were not available for Ankara, Bahçelievler. To obtain these data for the UBEM simulation, we applied a population downscaling strategy from an open source dataset [60]. In this approach, household size statistics from Ankara regional data are mapped to the analyzed area using figures from the national statistical database [61]. We calculated the density by dividing the household size by the floor area of each thermal zone. The resulting representative dataset is then randomly assigned to the thermal zone. To identify a general trend from the dataset, we used KDE and adjusted for all outliers outside the 95% confidence interval using a linear forward interpolation technique.

With NPDE, estimation errors may occur depending on the missingness mechanism of the observed data. Missingness mechanisms concern the relationship between missingness and the values of the variables in the data matrix. Rubin proposed three missingness categories: Missing Completely at Random (MCAR), Missing at Random (MAR) and Missing not at Random (MNAR) [44]. In MCAR, the missingness probability of a variable is completely independent of the variable and other variables. In MCAR, both single and multiple imputation methods are generally considered valid. Since we consider our case as MCAR, we apply multiple imputation by NPDE, in which the missing fields are randomly filled with new data. However, the probability of

missingness may also depend on observed data (for MAR) or unobserved data (for MNAR). In both cases, missingness occurs according to a certain pattern, so that an analysis of the joint distributions or modeling of the missingness mechanisms is required as part of the estimation procedure. NPDE can also be used with MAR, provided there is sufficient information to accurately estimate the conditional density of missingness from the observed data. However, NPDE is not suitable for MNAR because the estimates are biased due to the influence of unobserved variables. In this case, other imputation methods such as multiple imputation or maximum likelihood imputation can be used.

3.2. Data generation with parametric density estimation (PDE)

In our UBEM development case, the data of occupants (with the exception of people density), SHGC, $\text{COP}_{\text{Boiler}}$ and I_{rate} were completely missing. Since the complete lack of data does not allow us to make assumptions about the distribution, a viable approach is to use distributions defined in studies with similar building characteristics (i.e. building type, year of construction, climate) in the literature. These data sets are usually defined with PDE. In statistics, the probability density function $f(x)$ is used to describe the distribution of a random variable X as follows:

$$P(a < X < b) = \int_a^b f(x)dx \quad \text{for all } a < b \tag{1}$$

For continuous random variables, the standard distributions Uniform, Normal and Exponential, the three most common PDFs, form a basis for the many other distributions [62]. These standard distributions can be adjusted by changing their parameters to generate the desired distributions. Location and scale parameters are commonly used to define probability distribution functions as shown in Table 3. The location parameter shifts the graph on the x -axis, while the scale parameter stretches it with respect to the standard distributions mentioned above.

The data generation in our study was made with the standard distributions in reference to existing literature, which focuses on residential buildings (Table 4). As with NPDE, all parameters are considered independent. Uniform distributions, ensuring equal probability of occurrence within certain limits, were applied to a number of parameters including T_{H} , SHGC, $\text{COP}_{\text{Boiler}}$ and Q_{lighting} , $Q_{\text{equipment}}$ and I_{rate} . As there was no prior indication of a specific mean value or clustering for these parameters, a uniform approach was more suitable than a normal distribution, as it allowed the entire possible range to be represented with equal probability. The 'a' and 'b' values, which comprise the minimum and maximum values of the parameters, were carefully selected based on a synthesis of current research to ensure the robustness of the simulation in relation to the actual variability of the building and input data consistency. This choice supports consistency of industry norms and comparability between studies. The study's robust methodology relies on ASHRAE Standard 90.1-2013 for I_{rate} values. These parameters

Table 3
Summary of basic standard parametric distributions.

Distribution	Explanation	Probability distribution function (PDF)	
Uniform	A is the location parameter and (B - A) is the scale parameter. The case where A = 0 and B = 1 is called the standard uniform distribution.	$f(x) = \frac{1}{B - A} \text{ for } A \leq x \leq B$	(2)
Normal	μ is the location parameter and σ is the scale parameter. The case where $\mu = 0$ and $\sigma = 1$ is called the standard normal distribution.	$f(x) = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}$	(3)
Exponential	μ is the location parameter and β is the scale parameter (the scale parameter is often referred to as λ , which equals $1/\beta$). The case where $\mu = 0$ and $\beta = 1$ is the standard exponential distribution.	$f(x) = \frac{1}{\beta} e^{-(x-\mu)/\beta} x \geq \mu; \beta > 0$	(4)

Table 4
Data generated with standard parameter distributions.

Input parameter	Distribution			Reference
	Type	a	b	
T_{H}	U	20	24	[63]
SHGC	U	0.30	0.85	[37]
$\text{COP}_{\text{Boiler}}$	U	0.8	0.95	[29]
Q_{lighting}	U	12	19	[64]
$Q_{\text{equipment}}$	U	2.93	1.21	[65]
I_{rate}	U	0.000285 (ASHRAE 90.1-2013)	0.0005 (Leaky Building)	[27]

N = normal distribution, a = mean, b = standard deviation.
U = uniform distribution, a = min, b = max.

Table 5

The semantic data of the UBEM versions.

Building-related	V_{BASELINE}		V_{OCCUPANT}		$V_{\text{CONSTRUCTION}}$		V_{COMBINED}	
	Source	Value	Source	Value	Source	Value	Source	Value
WWR	—	0.2	—	0.2	—	NPDE	—	NPDE
U_{Wall} *	[69]	0.5	[69]	0.5	[69]	NPDE	[69]	NPDE
U_{Ground}	[69]	0.45	[69]	0.45	[69]	NPDE	[69]	NPDE
U_{Roof}	[69]	0.3	[69]	0.3	[69]	NPDE	[69]	NPDE
U_{Win}	[69]	2.4	[69]	2.4	[69]	NPDE	[69]	NPDE
SHGC	[70]	0.75	[70]	0.75	[70]	PDE	[70]	PDE
Infiltration Rate	[70]	2000: 0.000285	[70]	2000: 0.000285	[70]	PDE	[70]	PDE
		1980: 0.0004		1980: 0.0004				
		1960: 0.0005		1960: 0.0005				
$\text{COP}_{\text{Boiler}}$	[70]	0.85	[70]	0.85	[29]	PDE	[70]	PDE
Occupant-related	Source	Value	Source	Value	Source	Value	Source	Value
Q_{Peop}	[70]	0.028	[61]	NPDE	[70]	0.028	[61]	NPDE
Q_{Equip}	[70]	3.875	[65]	PDE	[70]	3.875	[65]	PDE
Q_{Light}	[70]	11.84	[64]	PDE	[70]	11.84	[64]	PDE
T_{H}	[70]	21.1	[46]	PDE	[70]	21.1	[46]	PDE

* NPDEs are calculated separately for concrete and masonry buildings.

are an essential component in representing the thermal performance of residential buildings and provide a solid foundation for UBEM's subsequent analysis and interpretation processes.

4. UBEM development and simulations

This chapter describes the development and simulation process of UBEM using the data sets described in the previous chapter. The energy models for buildings are developed down to the zone level. Modeling at the zone level also allows the use of occupant data typically associated with thermal zones and increases the reliability of performance results. Due to the high number of buildings in the region, the building energy simulation processes have been automated. EnergyPlus (V9.2.0) [66] and Honeybee [67] are used for building energy modeling and simulation. The Python library Seaborn is used for data analysis and visualization [68].

To demonstrate the impact of the previously presented data generation and augmentation techniques, we developed four different UBEM model versions based on the same 3D model but using different semantic data sets as shown in Table 5.

- V_{BASELINE} represents a case where semantic in-situ data is completely missing and the existing national standards or reference building definitions/libraries are used instead.
- $V_{\text{CONSTRUCTION}}$ is based on the same 3D model as V_{BASELINE} , but includes the newly generated building-related building data.
- V_{OCCUPANT} is based on the same 3D model as V_{BASELINE} , but includes the newly generated occupant-related semantic data.

- V_{COMBINED} combines V_{OCCUPANT} and $V_{\text{CONSTRUCTION}}$ and is the most complete model.

Modeling is performed in two main steps: geometric modeling and the assignment of semantic data to the geometric model. Geometric modeling includes the development of a 3D model of the buildings and the generation of thermal zones. Thermal zones are either the residential units or the unconditioned core spaces on each floor. The semantic data is divided into building-related and occupant-related features and integrated into the energy models in the second step. These two categories also differentiate the four UBEM model versions. While all UBEM versions are based on the same 3D model (buildings and zones), the versions differ from each other based on occupant- and building-related datasets. Finally, we present a comparative analysis between all UBEM versions in terms of energy demand (Q_{H}), global warming potential (GWP) and occupant comfort to highlight the impact of data detail levels on the simulation results.

4.1. 3D model development

4.1.1. 3D physical modeling and simulations

Fig. 4 illustrates the axonometric flowchart of the UBEM physical modeling process, highlighting both 3D modeling and thermal zoning. The first phase shows the simplified building footprints on the 2D master plans where the residential buildings are identified and assigned unique identifiers to each parcel and block. To optimize computational efficiency, the original layout drawings of buildings were simplified into basic four-corner rectangles. Additionally, 10% of the total parcel area

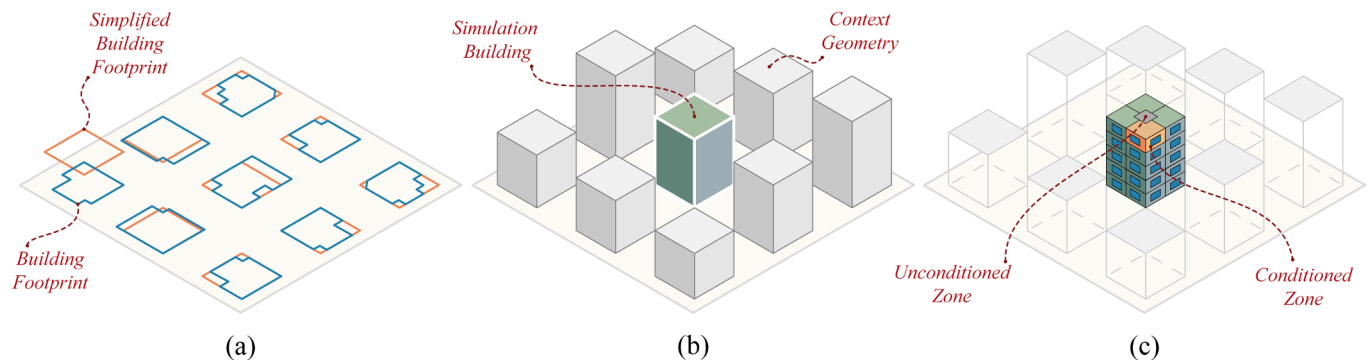


Fig. 4. Axonometric flowchart of physical modeling and simulation process of UBEM, (a) Geometry simplifications, (b) Building level 3D physical modeling, (c) Residential thermal zoning.

was allocated to vertical circulation areas.

The second phase shows the 3D mass and context modeling. Using the data on the number of floors, we constructed 3D volumes and performed separate energy simulation steps for each building where the surrounding buildings within a radius of 20 m are taken into account to account for shading and interactions with urban heat during the sequential energy simulations of the residential buildings as shown in Fig. 4.

Finally, the third phase describes the process of thermal zoning as the buildings are divided into floors as thermal zones. We created two types of thermal zones: (1) conditioned occupied residential zones for residential areas, (2) unconditioned building cores. The residential buildings were simulated at the zone level, while the remaining commercial and public buildings were not simulated during the UBEM process. Attached buildings were modeled separately, with intersecting walls treated as adiabatic surfaces.

To streamline the simulation process, we developed an automated model to simulate each building individually and sequentially for the entire neighborhood using the Grasshopper tool inside the Rhinoceros 3D modeling program [71]. Grasshopper was chosen due to its algorithmic modeling capabilities. It allows for automated simulations of multiple buildings in the neighborhood with parametric control. Grasshopper integrates with Ladybug Tools [67] that consist of EnergyPlus (E+) [66] which is enabling comprehensive building energy simulations within the workflow.

4.1.2. Assigning semantic data

We introduce semantic data in two phases: In the first phase, we assign occupant-related data to residential zones, and in the second phase, we assign construction-related data to buildings. We only assign occupant-related data to residential zones, as the core zones are considered unoccupied. We assume uniform occupancy characteristics for V_{BASELINE} and $V_{\text{CONSTRUCTION}}$, assigning to residential zones the datasets defined in the U.S. Department of Energy's (DOE) 'Multi-family low-rise apartment building' typology [72]. In V_{OCCUPANT} and V_{COMBINED} , a different household with different occupancy characteristics is considered for each zone. In V_{OCCUPANT} and V_{COMBINED} ; therefore, we use data sets that we previously created with PDE and NPDE for Q_{Equip} , Q_{Light} , T_{H} and Q_{Peop} .

We assign building-related data to each building. For masonry and concrete buildings, we specify all materials separately for wall, roof and ground floor constructions to account for their thermal mass. For this,

we used the typical construction sets proposed in a similar study in Ankara (Table 6) [73]. On the other hand, surface construction data is available in the EPC datasets for both building types. In V_{BASELINE} and V_{OCCUPANT} , we used the U-values specified in the current Turkish construction standard for all buildings [69]. In $V_{\text{CONSTRUCTION}}$ and V_{COMBINED} , we first generate U-values with NPDE for each building. We then vary the insulation thickness to obtain the necessary U-value for that building. However, in the V_{BASELINE} and V_{OCCUPANT} versions, the construction type is ignored and the U-values are generated according to the Turkish standards.

Other model parameters that apply to all buildings are as follows: The cooling demand is met by natural ventilation, where windows are opened as soon as the indoor air temperature is above 24 °C. Mechanical cooling is ignored. The schedules for occupancy, lighting and equipment are also used based on 'Multi-family low-rise apartment building' typology [72]. Finally, the setpoints for heating and cooling are derived from the values given in the literature [74].

4.2. Performance objectives

We have calculated the performance objectives using the UBEM simulation results for each residential zone. The energy demand for heating (Q_{H}) is based on the living area (kWh/m^2), although cooling energy is not included in the simulations as the buildings are not air-conditioned. We have quantified the energy demand for lighting (Q_{L}) and equipment (Q_{E}) in a similar way to Q_{H} . The carbon footprint of the zones is expressed as Global Warming Potential (GWP) per floor area ($\text{kg CO}_2\text{-eq/m}^2$). We estimated GWP values using energy demand data for all energy demand types. To calculate the GWP for Q_{H} , we used the average GWP for natural gas boilers of $0.181 \text{ kg CO}_2\text{-eq/m}^2$ [75], which is consistent with GWP values from related studies [51,53,76]. To calculate the GWP for Q_{L} and Q_{E} , we used grid electricity with a Turkish grid GWP of $0.545 \text{ kg CO}_2\text{-eq/m}^2$, as reported in [77] and confirmed by life cycle analyses of Turkish electricity generation [78].

$$T_n = 0.31T_{\text{ave}} + 17.8 \quad (7)$$

$$T_{\text{conf},i,z} = T_n + 2.5 \quad (8)$$

$$IOD = \frac{\sum_{i=1}^{N_{\text{occ}(z)}} [\max\{OT_{i,z} - T_{\text{conf},i,z}, 0\} \times t_{i,z}]}{\sum_{i=1}^{N_{\text{occ}(z)}} t_{i,z}} \quad (9)$$

Table 6

V_{BASELINE} construction properties [73].

Construction	Material	thickness (m)	conductivity (W/m-K)	density (kg/m^3)	specific heat (J/kg-K)
External wall, masonry	External plaster	0.02	0.721	1762	840
	EPS	0.05	0.035	22	1400
	Brick, solid	0.2	0.5	1200	800
	Internal plaster	0.02	0.51	1200	840
External wall, concrete	External plaster	0.03	1.6	2000	840
	EPS	0.0477	0.035	22	1500
	Brick, perforated	0.135	0.33	600	900
	Internal plaster	0.02	0.51	1200	1090
Roof	Metal roofing	0.0015	45.006	7680	418.4
	Glass wool	0.121	0.04	18	670
	Reinforced-Concrete	0.12	2.5	2400	860
	Plaster	0.02	0.51	1200	840
Ground floor	Gravel	0.1	2.0	2000	840
	Concrete	0.1	1.65	2200	860
	Insulation, water	0.003	0.19	3000	780
	Concrete	0.1	1.65	2200	860
	XPS	0.058	0.035	25	1200
	Polyester	0.003	0.19	25	1500
	Screed	0.05	1.4	2000	837
	Flooring, laminate	0.008	0.115	938	1600
Internal wall	Plaster	0.02	0.51	1200	1090
	Brick, perforated	0.085	0.33	600	900
	Plaster	0.02	0.51	1200	1090

Indoor Overheating Degrees (IOD) are used as a measure to assess the thermal load in naturally ventilated buildings during the summer months (May 30 – September 1). The employed method takes into account both the frequency of overheating and the intensity of overheating — the latter being the degree to which indoor temperatures exceed a predetermined upper comfort threshold [79]. First, we determined the comfort neutrality temperature (T_n) according to Eq. (7), relying on the monthly average outdoor air temperature (T_{ave}) according to ASHRAE 55-2013 standards [80]. We then used Eq. (8) to determine the upper limit of thermal comfort ($T_{comf,i,z}$) for each zone during occupied periods. The IOD calculation is then performed using Eq. (9), which takes into account the total number of occupied hours in zone z (i.e. $N_{occ}(Z)$) for each hour and the operative temperature in each hourly interval ($OT_{i,z}$).

5. Results

The comprehensive evaluation presented in this study highlights the intricate relationships between UBEM's inputs and performance outcomes, thereby revealing potential ways to optimize reliability and efficiency through data imputation and generation methods used in bottom-up UBEM processes. The study compares four different

iterations of UBEM in detail: $V_{BASELINE}$, $V_{OCCUPANT}$, $V_{CONSTRUCTION}$, $V_{COMBINED}$. The diversity of performance results is explicitly analyzed in both visual and written form.

5.1. Correlation analysis between inputs and performance objectives

This section examines the relationships between different inputs and performance targets in the context of the UBEM using a correlation matrix and facilitates a comprehensive understanding of the correlations between the inputs and the key performance indicators GWP, $\dot{Q}_H$ and IOD. During data generation, each input was randomly generated based on unique seeds to ensure that there are no correlations between the generated parameters. Consequently, inputs that do not vary between versions are not included in the matrix as they have no correlation. The distinction between semantic variables in versions is illustrated, in Fig. 5. In the appendix section, additionally, we presented extended correlation analysis visualizations for all UBEM versions, as Figs. B1-B4.

Among the geometric inputs, only the window-to-wall ratio (WWR) varies between the UBEM versions. The infiltration rate is the only input that differs in all versions. However, the parameter shows a weak correlation with the three performance objectives. WWR_{east}, U_{ground} , U_{window} and T_H show no distinct correlation with the performance objectives in all versions. The inputs COP_{boiler} and Q_{people} show a negative



Fig. 5. Correlation matrices of UBEM versions.

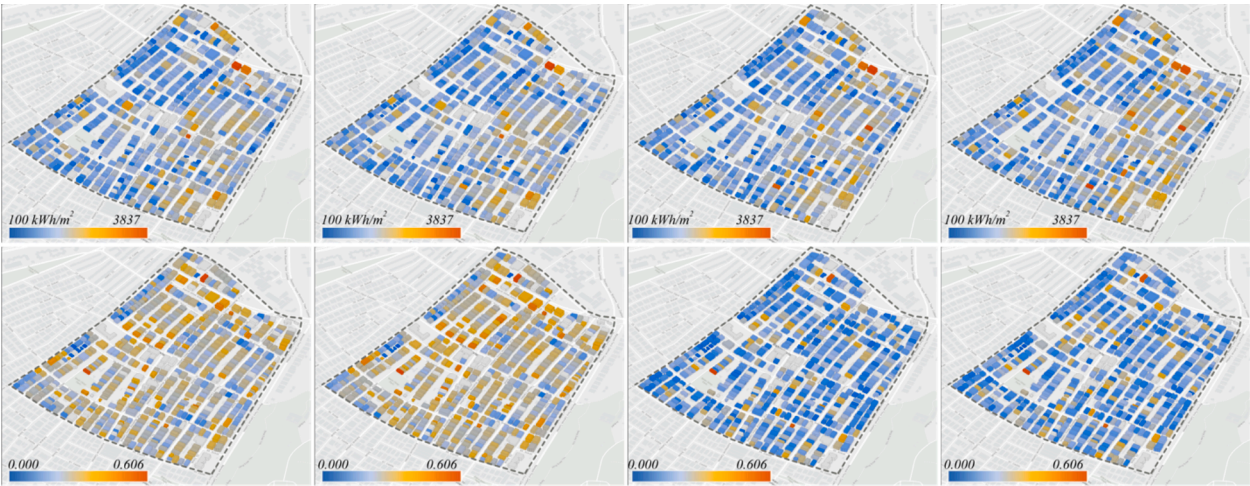


Fig. 6. Building-based performance objective for UBEM versions; (up) Q_H , (a) $V_{BASELINE}$, (b) $V_{OCCUPANT}$, (c) $V_{CONSTRUCTION}$, (d) $V_{COMBINED}$; (bottom) IOD, (a) $V_{BASELINE}$, (b) $V_{OCCUPANT}$, (c) $V_{CONSTRUCTION}$, (d) $V_{COMBINED}$;

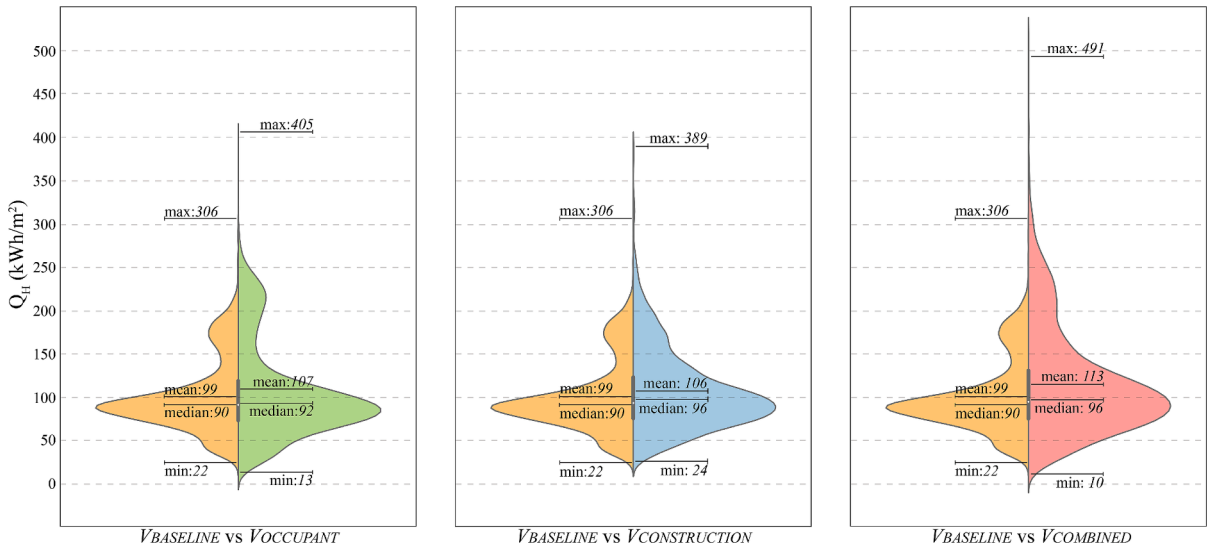


Fig. 7. Heating demand (Q_H) UBEM versions, violin plot.

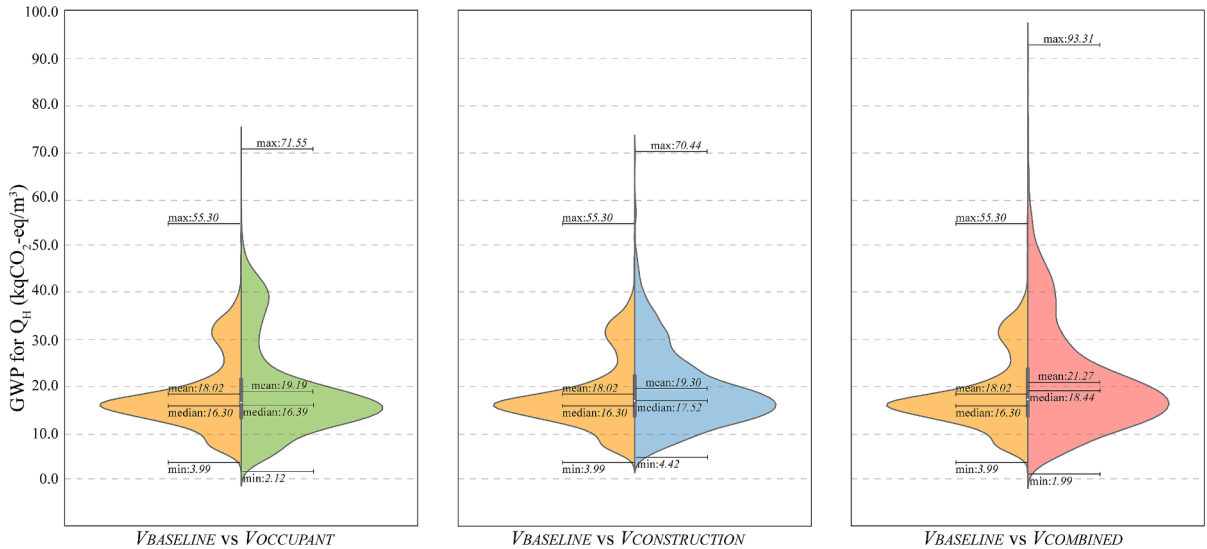


Fig. 8. Global warming potential for Q_H UBEM versions, violin plot.

correlation with GWP and Q_H , while the other inputs exhibit low positive correlation. In particular, the inputs WWR_{south} and $SHGC_{\text{window}}$ show a high correlation with the IOD.

5.2. Building-based performance objectives

In this section, spatial performance comparisons for Q_H and IOD are evaluated from a building-based axonometric perspective (Fig. 6). IOD is determined by averaging the performance data of the residential zones within the buildings, while Q_H is calculated by summing the results. The application of randomly generated distributions results in a reduction of the overall Q_H objectives for buildings, which is visually represented by a shift towards blue. In particular, the distinct color differences between V_{BASELINE} and the $V_{\text{CONSTRUCTION}}$ and V_{COMBINED} versions highlight the transition to blue, indicating a lower energy demand. Despite these changes, some buildings with high energy requirements show similar IOD values across the different versions, indicating consistent performance in these areas. Overall, Q_H generally shows no significant differences between versions, suggesting that while building inputs affect IOD, their influence on Q_H is rather limited.

The comparisons of energy balance components that are added in Appendix D. Fig. D.1 illustrate the gains and losses of all UBEM versions as density charts, separated by demand objectives, including conduction, equipment demand, lighting demand, mechanical ventilation, natural ventilation, solar gain, and people gain. Additionally, Fig. D.2 presents the average energy balance of all versions for the same setup. It highlights the distribution of energy demand components across UBEM versions.

5.3. Zone-based performance objectives

Fig. 7 shows three performance objectives for each version in comparison with the violin plot, which visualizes the data distribution in terms of density. All results are based on the average of the annual Q_H for residential zones per m^2 .

The Q_H of the residential buildings of Bahçelievler showed close values for all four versions in terms of μ (mean) and η (median), i.e. μ is between 99 and 113 kWh/m^2 , η is between 90 and 96 kWh/m^2 . All three versions were more widely distributed than V_{BASELINE} . While the V_{OCCUPANT} and V_{COMBINED} versions have a double-peaked shape similar to V_{BASELINE} , the results in the $V_{\text{CONSTRUCTION}}$ version are concentrated under a single peak. The version with the most extreme outliers was V_{COMBINED} , which had the widest range in both minimum and maximum

values, e.g. 10 and 491 kWh/m^2 . V_{COMBINED} has the most varied inputs compared to the other versions.

For GWP violin plot visualizations as shown in Fig. 8, the results are GWP for annual Q_H for per $kg\ CO_2\text{-eq}/m^2$. The Q_H of Bahçelievler residential buildings resulted in close μ and η values for all four versions, i.e. μ is between 18.02 and 21.27 $kg\ CO_2\text{-eq}/m^2$, η is between $kg\ CO_2\text{-eq}/m^2$. All three versions were more widespread than V_{BASELINE} . While the V_{OCCUPANT} and V_{COMBINED} versions have a two-peaked shape similar to V_{BASELINE} , the results in the $V_{\text{CONSTRUCTION}}$ version are concentrated under a single peak. The version with the most extreme outliers was V_{COMBINED} , which showed the widest range for both minimum and maximum values, e.g. 10 and 491 $kg\ CO_2\text{-eq}/m^2$.

For IOD violin plot visualizations as shown in Fig. 9, the results are the average of the annual indoor overheating degrees (IOD) for all zones. The residential zones are thus simulated as a whole volume; no room division was made in any version. The IOD resulted in close values in terms of μ and median values for V_{BASELINE} and V_{OCCUPANT} , i.e. μ is between 0.1973 and 0.2147, η is between 0.1971 and 0.2125. The IOD revealed close μ and η values for $V_{\text{CONSTRUCTION}}$ and V_{COMBINED} , i.e. μ is between 0.1394 and 0.1579, η is between 0.1143 and 0.1326. μ and η values were lower for $V_{\text{CONSTRUCTION}}$ and V_{COMBINED} . In addition, the distribution of these two versions can also be seen in the violin diagram, concentrated at lower values. On the other hand, the extreme values in both versions are significantly higher than V_{BASELINE} , namely 0.7158 and 0.8165.

Lastly, in the appendix section, Table A.1 provides a statistical description of all UBEM versions in terms of Q_H , IOD, and GWP performance objectives. Also, Fig. C.1 shows the Density Chart for all versions in terms of Q_H and IOD.

Table 7
Energy balance of versions, average.

kWh/m^2	V_{BASELINE}	V_{OCCUPANT}	$V_{\text{CONSTRUCTION}}$	V_{COMBINED}
$Q_{\text{Conduction}}$	-70.42	-74.58	83.46	91.42
$Q_{\text{Natural Ventilation}}$	-19.53	-23.77	-16.38	-21.17
$Q_{\text{Mechanical Ventilation}}$	-53.17	-55.64	-52.78	-55.29
$Q_{\text{Infiltration}}$	-51.26	-54.08	-43.76	-46.22
Q_{People}	20.36	23.36	20.36	23.36
Q_{Lighting}	34.06	44.45	34.06	44.45
$Q_{\text{Equipment}}$	22.33	17.16	22.33	17.16
Q_{Solar}	33.00	33.00	27.27	27.27
Q_{Heating}	84.63	90.10	92.36	101.86

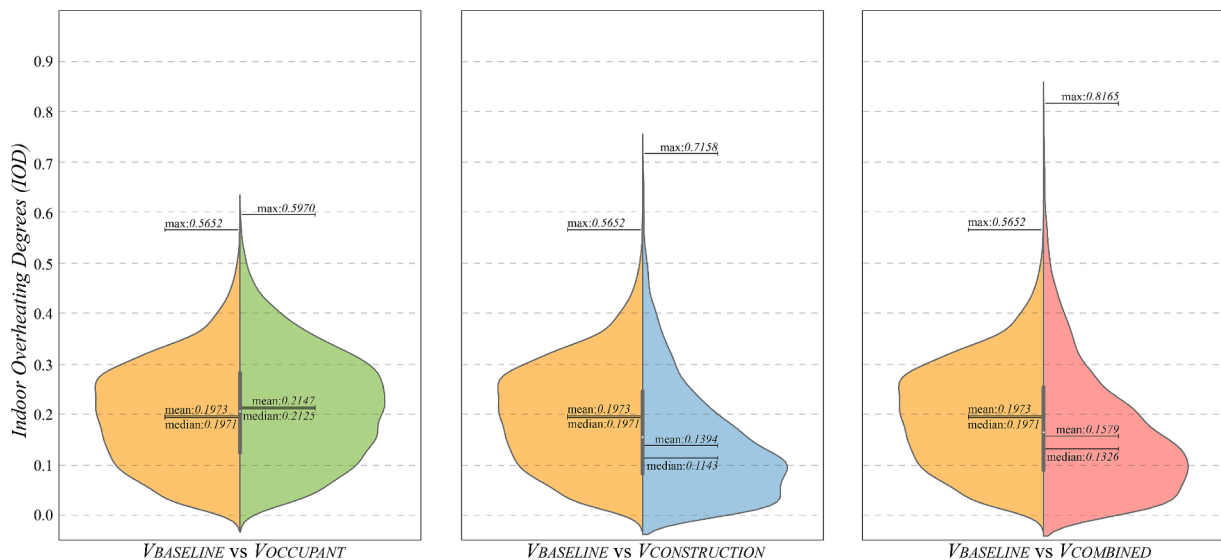


Fig. 9. Overheating degrees UBEM versions (IOD), violin plot.

5.4. Energy gains and losses of all versions

For the energy gains in kWh/m², there were different results between the versions in terms of solar gain, lighting and heating loads. The solar gains of V_{BASELINE} and V_{OCCUPANT} are higher than the other two versions. The lighting loads were higher for V_{OCCUPANT} and V_{COMBINED} than for the other versions. The heating loads were higher for V_{COMBINED} than for the other versions. In terms of energy losses in kWh/m², significant differences were found in the loads for conduction of all internal and external surfaces, infiltration rate and natural ventilation. The value of conduction losses was higher for the other versions than for V_{BASELINE}. The energy loss due to infiltration is high for V_{BASELINE} and V_{OCCUPANT}. The energy loss due to natural ventilation is significantly higher for V_{OCCUPANT} and V_{COMBINED} than for the other versions.

Table 7 contains a numerical representation of the average kWh/m² for residential zones. It describes the energy balance of four UBEM versions: V_{BASELINE}, V_{OCCUPANT}, V_{CONSTRUCTION} and V_{COMBINED}. In these versions, Q_{Conduction} shows a significant increase from V_{BASELINE} to V_{COMBINED}, indicating improved thermal performance. Q_{NaturalVentilation} shows a decrease in energy losses in all versions except V_{CONSTRUCTION}, indicating different ventilation strategies. The losses from mechanical ventilation and infiltration are relatively consistent, with V_{CONSTRUCTION} showing the lowest energy loss from infiltration. The contributions from occupants, lighting and appliances increase in the V_{OCCUPANT} model. This reflects the impact of occupancy patterns on energy demand. Q_{Solar} remains constant for V_{BASELINE} and V_{OCCUPANT} but decreases for V_{CONSTRUCTION} and V_{COMBINED}. It is noteworthy that Q_{Heating} increases from V_{BASELINE} to V_{COMBINED}, which illustrates the increased Q_H in the combined model. These tables illustrate the dynamic interaction of building components and occupant behavior in influencing the energy profile of the buildings studied.

5.5. Key influences on UBEM performance

We performed a Random Forest Feature Importance analysis for Q_H, IOD and GWP to assess the impact of generated input parameters on the performance of UBEM. Since V_{COMBINED} includes all parameter variations, we applied the analysis to this version. The results show that the building form and the thermal envelope properties are the most important factors influencing the energy performance.

For Q_H, we found that the form factor dominates (70.3%), which emphasizes the role of building geometry and the surface-to-volume ratio in heat loss. The thermal envelope properties (wall and roof U-values ~11%) also play a significant role. However, occupant-related factors (e.g., equipment, lighting and occupant density) make a minimal contribution (~1–1.5%), suggesting that internal heat gains have a smaller impact on the heating load. For IOD, our analysis shows that solar-related parameters determine the risk of overheating. SHGC (31.3%), vertical position (24.0%) and south-facing WWR (12.4%) have a strong influence on overheating. For GWP, we observed that form factor remains the dominant factor (64.0%), followed by equipment density (6.6%) and lighting density (5.6%). The influence of plug loads and lighting are effective on emissions. In contrast to IOD, solar exposure plays a minor role (~1%), suggesting that energy demand and system efficiency have a greater influence on emissions than solar heat gains.

Three important findings can be derived from these results:

- Building form and building envelope performance are the strongest determinants of UBEM accuracy, highlighting the need for high-resolution geometric and material data.
- Solar exposure has a significant impact on IOD, but only limited influence on heating demand and emissions.
- Occupant-related factors have a moderate impact (~2–6%), confirming that physical building characteristics are more important than operational impacts in UBEM simulations.

5.6. Discussion

In this study, we created four versions of UBEM to demonstrate the impact of data generation and augmentation techniques on the reliability of the simulation. V_{BASELINE}, V_{CONSTRUCTION}, V_{OCCUPANT}, and V_{COMBINED} versions were developed using the same 3D model but incorporated different semantic data. The main results showed significant performance differences between these versions. V_{COMBINED} exhibited the greatest variability in key metrics, i.e. Q_H, GWP and IOD. This highlights the value of comprehensive data integration. Correlation analysis indicated that certain parameters, e.g. WWR and infiltration rate, varied between models and affected performance objectives differently. For instance, the inputs COP_{Boiler} and Q_{People} correlated negatively with GWP and Q_H, while other inputs had low positive correlations. Statistical analyses showed that the mean and standard deviations for Q_H and GWP were higher in the models that included occupant and construction data. V_{COMBINED} had the highest correlation values, demonstrating improved precision and detail. Violin plots further illustrated these variations, with V_{COMBINED} showing the widest range of values, indicating its ability to capture extreme cases. Analysis of the energy balance showed that energy gains and losses varied significantly between versions. For example, solar gains were higher in V_{BASELINE} and V_{OCCUPANT}, while V_{COMBINED} had higher heating loads, reflecting the impact of comprehensive data on energy demand. These results emphasize the importance of gradual data integration in developing accurate and reliable UBEMs that can help in decision making to address the impacts of climate change.

Nevertheless, there were some limitations to this study in terms of data availability and integration. For UBEM, there is no publicly available occupancy data for residential buildings; however, the available data is a statistical analysis rather than raw, publicly available datasets. For building data, coverage is incomplete for all buildings and some outliers had to be cleaned for robust statistical results. Geometric data — such as building layout, number of floors and total residential units per building — is collected by different government agencies, including the Ankara Metropolitan Municipality and the Ministry of Environment, Urbanization and Climate Change. Data sharing has been restricted due to the Personal Data Protection Law. In addition, public geometric data originally collected for tax and land transactions had to be transformed before it was suitable for energy modeling. More transparent and coordinated data sharing between government agencies would enable the development of a comprehensive, nationwide UBEM database with greater detail and reliability. This study was conducted to demonstrate that the development of a UBEM database is feasible despite the scarcity of data.

6. Conclusion

UBEM is in a new approach of development, where there is a lack of standardized methods, which is impacting the reliability of reported accuracies. The lack of standardized practices in terms of spatial and temporal resolutions and error measurements makes it difficult to compare the results of different studies. To advance this field, it is essential to introduce transparent methods that systematically report the robustness of UBEM at different resolutions. However, a key challenge in UBEM is the lack of complete and reliable data sets, which hinders the development of valid energy models, especially in data-scarce environments. Therefore, this study approached the problem of data scarcity problem by employing a novel approximation method using statistical methods to generate missing data. Imputation of data was performed using locally sourced open datasets or by generating data referencing global resources when similar local datasets were not available. This methodological approach provides a pathway for constructing practical UBEMs even with limited data.

We have significantly increased the resolution of UBEM down to the zone level. This is crucial for improving the resolution of the simulations

compared to typical UBEM studies that focus on building or floor level analysis. By applying different versions of UBEM during data generation and implementation, we examined the effects of different parameter groups, which allowed us to determine which parameters most significantly influence model reliability of the model.

The quantitative results also illustrate the impact of the different UBEM versions on the performance results. As can be seen from Fig. 7, Fig. 8 and Fig. 9, the variability in heating demand between UBEM versions is considerable. Specifically, the heating demand ranges from 22 to 306 kWh/m² for V_{BASELINE} and from 10 to 491 kWh/m² for V_{COMBINED} . The global warming potential (GWP) also varies between 3.99 and 55.30 kg CO₂-eq/m² for V_{BASELINE} and between 1.99 and 93.31 kg CO₂-eq/m² for V_{COMBINED} . These results highlight the significant influence of data generation approaches on the UBEM results and emphasize the need for high-resolution models to capture the full range of performance variability within urban-scale energy simulations.

The strengths of our study include obtaining physical characteristics of building and residential datasets from publicly available sources. We also generated occupant-related data using both parametric (PDE) and non-parametric (NPDE) methods to compensate for the lack of available occupant data in the region of analysis. The resulting distribution is unimodal, with a peak at 0.015, a mean of about 0.02 and a right skew that exceeds 0.03, indicating the presence of a small subset of rooms with relatively high occupancy density.

The availability of EPC data enabled the integration of different types of data from a single source, e.g. the number of buildings, block-parcel numbers, the total number of residential units and surface characteristics. Importantly, while most UBEM studies operate at the building level due to the vast number of buildings, our study improves the resolution by modeling at the residential unit level. This adjustment significantly improves performance by providing a more detailed representation of heat transfer and HVAC system use within residential units.

Our study encounters limitations regarding the completeness of the data. Our approach assumes Missing Completely at Random (MCAR) conditions. Under this assumption, we apply multiple imputation by Non-Parametric Density Estimation (NPDE), in which the missing fields are randomly filled with new data. This assumption could affect the veracity of the data imputation outcomes, potentially impacting the overall accuracy of the model.

Finally, future research should focus on the use of transfer learning

techniques, where models are first trained on a subset of data — either from a part of the target city or another city — and subsequently retraining them to fit the broader city dataset. This approach could significantly improve the scalability and applicability of UBEM methods in different urban contexts.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used instatext.io in order to check grammar, spelling. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

Orçun Koral İşeri: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Ayça Duran:** Writing – original draft, Methodology, Data curation. **İlkin Canlı:** Resources, Methodology, Formal analysis, Data curation, Conceptualization, Writing – original draft. **Çağla Meral Akgül:** Supervision, Data curation, Methodology, Formal analysis. **Sinan Kalkan:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **İpek Gürsel Dino:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Acknowledgements

The study was funded by the Scientific and Technological Research Council of Türkiye (TUBITAK) under the grant number 120M997.

Appendix A. — Zone-based performance objectives

Table A.1

Statistical Description of all Versions.

	V_{BASELINE}			V_{OCCUPANT}			$V_{\text{CONSTRUCTION}}$			V_{COMBINED}		
	Q_H	IOD	GWP	Q_H	IOD	GWP	Q_H	IOD	GWP	Q_H	IOD	GWP
M	99.57	0.20	48.75	106.00	0.21	52.75	106.61	0.14	50.02	117.54	0.16	54.84
std	41.30	0.10	7.47	56.14	0.11	10.84	46.67	0.11	8.46	63.62	0.12	12.11
min	22.02	0.00	34.72	11.73	0.00	0.00	24.41	0.00	0.00	11.00	0.20	0.00
max	305.5	0.57	44.15	395.32	0.60	106.73	389.15	0.72	101.67	515.50	0.82	128.48
Change	—	—	—	6.5 %	8.8 %	8.2 %	7.1 %	−29.3 %	2.6 %	18 %	−20 %	12.5 %

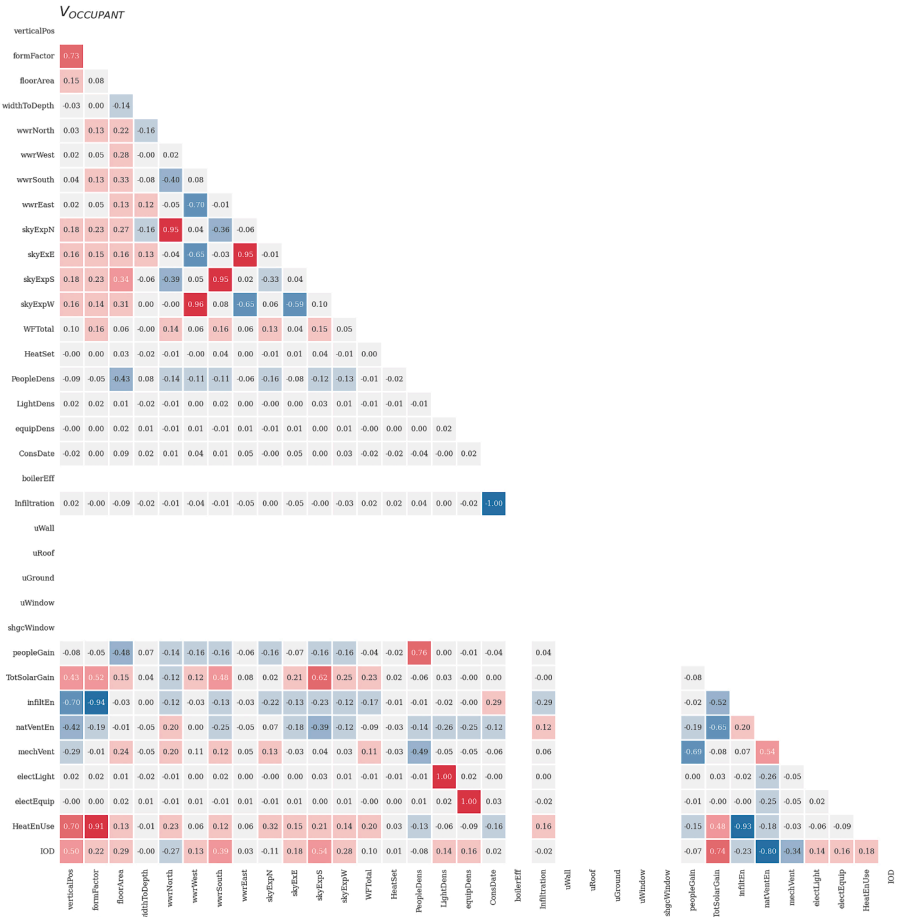


Fig. B.2. Heat Map of V_OCCUPANT



Fig. B.3. Heat Map of VCONSTRUCTION

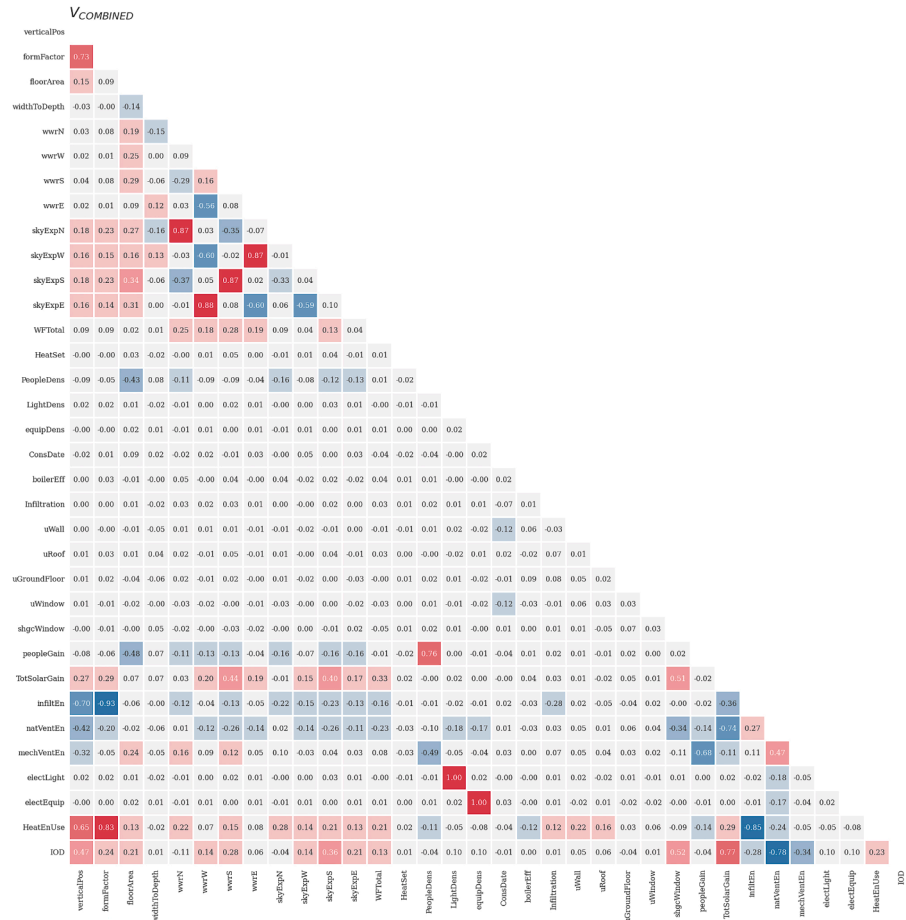


Fig. B.4. Heat Map of $V_{COMBINED}$

Appendix C. Simulation outputs, density charts

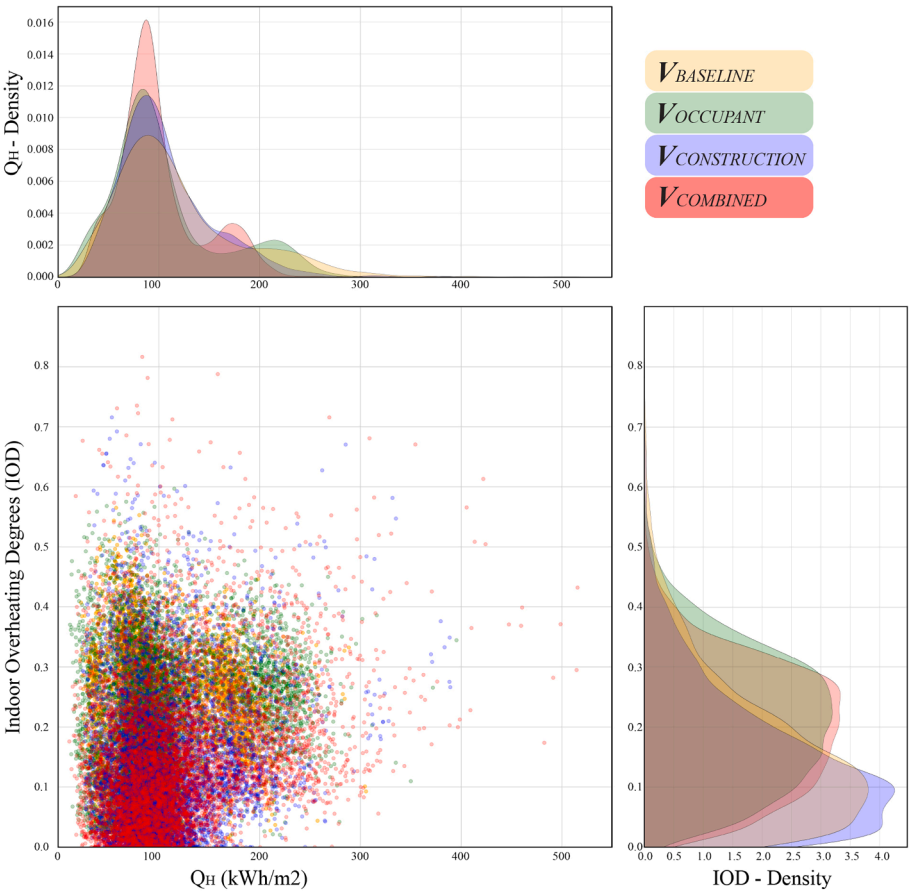


Fig. C.1. Density Chart for UBEM Versions in terms of Q_H and IOD

Appendix D. Comparisons of energy balance components

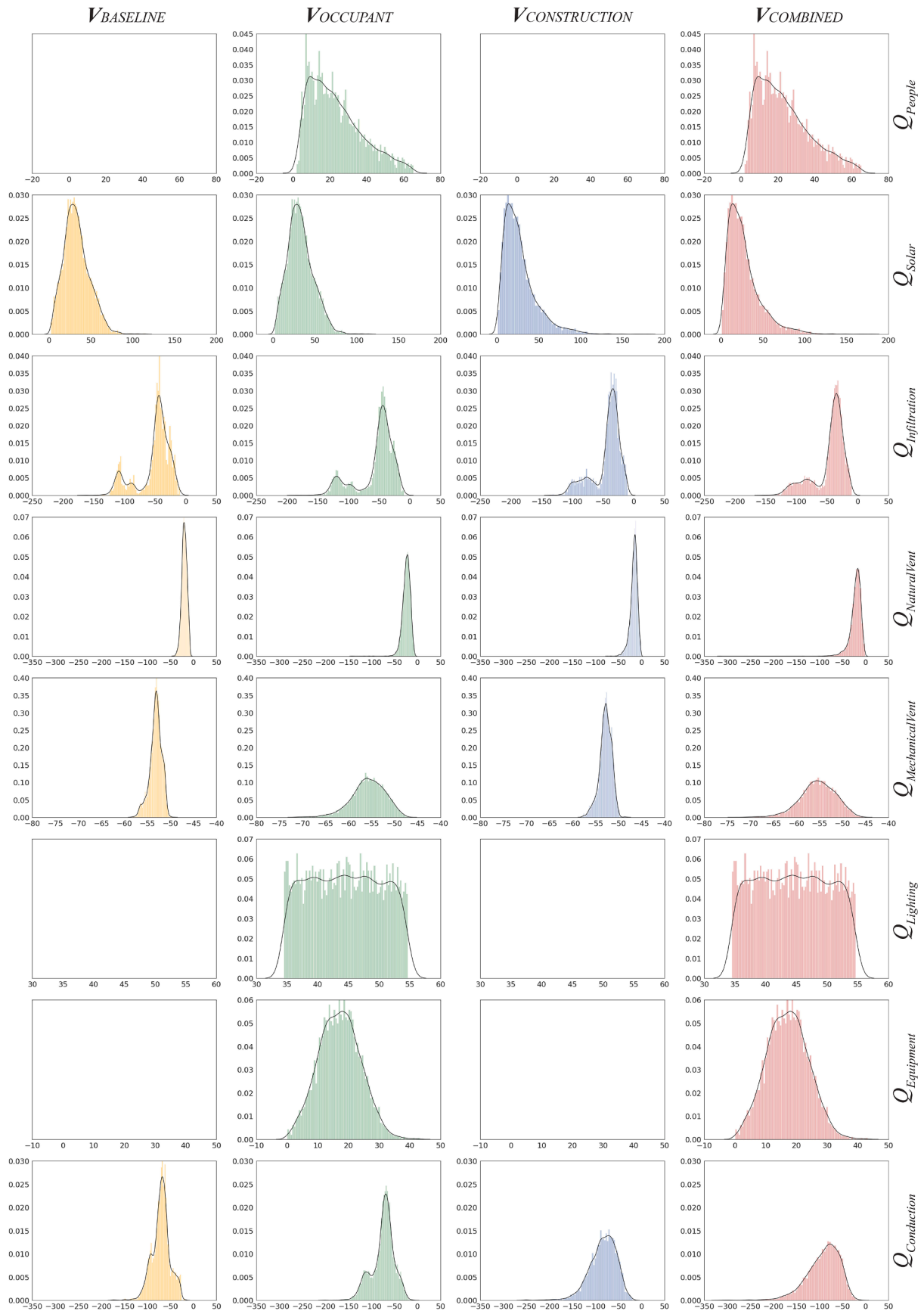


Fig. D.1. Gains and Losses of Versions

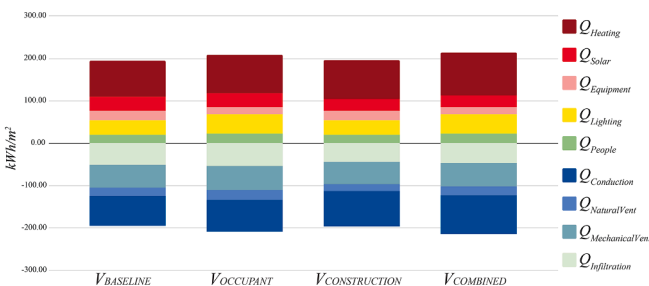


Fig. D.2. Energy Balance of Versions (Average)

Appendix E. Descriptive statistics of the versions

Table E.1
Descriptive Statistics of V_{BASELINE}.

Index	Qpeople	Qsolar	Qinfiltration	Qnatvent	Qmechvent	Qlight	Qequip	Qconduction	Qheat
count	6458	6458	6458	6458	6458	6458	6458	6458	6458
mean	23.36	27.27	-46.22	-21.17	-55.29	44.45	17.16	-91.42	101.86
std	14.24	19.59	25.98	12.31	3.83	5.78	6.78	35.96	54.49
min	1.87	0.89	-154.35	-315.49	-72.89	34.51	0.06	-315.41	9.27
0.25	11.87	13.34	-50.14	-26.01	-57.65	39.43	12.33	-111.92	64.76
0.50	20.71	22.63	-37.77	-19.05	-55.20	44.49	17.12	-85.95	88.63
0.75	31.93	35.06	-29.73	-13.43	-52.57	49.36	21.73	-65.13	124.49
max	65.23	178.91	-9.97	-0.96	-45.52	54.65	43.17	-10.89	477.87

Table E.2
Descriptive statistics of V_{OCCUPANT}.

Index	Qpeople	Qsolar	Qinfiltration	Qnatvent	Qmechvent	Qlight	Qequip	Qconduction	Qheat
count	6458	6458	6458	6458	6458	6458	6458	6458	6458
mean	23.36	27.27	-46.22	-21.17	-55.29	44.45	17.16	-91.42	101.86
std	14.24	19.59	25.98	12.31	3.83	5.78	6.78	35.96	54.49
min	1.87	0.89	-154.35	-315.49	-72.89	34.51	0.06	-315.41	9.27
0.25	11.87	13.34	-50.14	-26.01	-57.65	39.43	12.33	-111.92	64.76
0.50	20.71	22.63	-37.77	-19.05	-55.20	44.49	17.12	-85.95	88.63
0.75	31.93	35.06	-29.73	-13.43	-52.57	49.36	21.73	-65.13	124.49
max	65.23	178.91	-9.97	-0.96	-45.52	54.65	43.17	-10.89	477.87

Table E.3
Descriptive statistics of V_{CONSTRUCTION}.

Index	Qpeople	Qsolar	Qinfiltration	Qnatvent	Qmechvent	Qlight	Qequip	Qconduction	Qheat
Count	6458	6458	6458	6458	6458	6458	6458	6458	6458
mean	23.36	27.27	-46.22	-21.17	-55.29	44.45	17.16	-91.42	101.86
std	14.24	19.59	25.98	12.31	3.83	5.78	6.78	35.96	54.49
min	1.87	0.89	-154.35	-315.49	-72.89	34.51	0.06	-315.41	9.27
0.25	11.87	13.34	-50.14	-26.01	-57.65	39.43	12.33	-111.92	64.76
0.50	20.71	22.63	-37.77	-19.05	-55.20	44.49	17.12	-85.95	88.63
0.75	31.93	35.06	-29.73	-13.43	-52.57	49.36	21.73	-65.13	124.49
max	65.23	178.91	-9.97	-0.96	-45.52	54.65	43.17	-10.89	477.87

Table E.4

Descriptive statistics of V_{COMBINED}.

Index	Qpeople	Qsolar	Qinfiltration	Qnatvent	Qmechvent	Qlight	Qequip	Qconduction	Qheat
count	6458	6458	6458	6458	6458	6458	6458	6458	6458
mean	23.36	27.27	−46.22	−21.17	−55.29	44.45	17.16	−91.42	101.86
std	14.24	19.59	25.98	12.31	3.83	5.78	6.78	35.96	54.49
min	1.87	0.89	−154.35	−315.49	−72.89	34.51	0.06	−315.41	9.27
0.25	11.87	13.34	−50.14	−26.01	−57.65	39.43	12.33	−111.92	64.76
0.50	20.71	22.63	−37.77	−19.05	−55.20	44.49	17.12	−85.95	88.63
0.75	31.93	35.06	−29.73	−13.43	−52.57	49.36	21.73	−65.13	124.49
max	65.23	178.91	−9.97	−0.96	−45.52	54.65	43.17	−10.89	477.87

Data availability

Data will be made available on request.

References

- [1] M. Kirk, The global report on human settlements, Popul. Stud. (NY) 42 (1988) 519–520, <https://doi.org/10.1080/0032472031000143786>.
- [2] N. United, World Urbanization Prospects, 2018.
- [3] International Energy Agency, Transition to sustainable buildings, strategies and opportunities to 2050, 2013. doi:10.1787/9789264202955-en.
- [4] N. Buckley, G. Mills, C. Reinhart, Z.M. Berzolla, Using urban building energy modelling (UBEM) to support the new European Union's Green Deal: case study of Dublin Ireland, Energy Build. 247 (2021) 111115, <https://doi.org/10.1016/j.enbuild.2021.111115>.
- [5] F. Johari, G. Peronato, P. Sadeghian, X. Zhao, J. Widén, Urban building energy modeling: state of the art and future prospects, Renew. Sustain. Energy Rev. 128 (2020), <https://doi.org/10.1016/j.rser.2020.109902>.
- [6] S. Torabi Moghadam, C. Delmastro, S.P. Corngati, P. Lombardi, Urban energy planning procedure for sustainable development in the built environment: A review of available spatial approaches, J. Clean. Prod. 165 (2017) 811–827, <https://doi.org/10.1016/j.jclepro.2017.07.142>.
- [7] T. Hong, Y. Chen, X. Luo, N. Luo, S.H. Lee, Ten questions on urban building energy modeling, Build. Environ. 168 (2020) 106508, <https://doi.org/10.1016/j.buildenv.2019.106508>.
- [8] W. Wang, K. Liu, M. Zhang, Y. Shen, R. Jing, X. Xu, From simulation to data-driven approach: a framework of integrating urban morphology to low-energy urban design, Renew. Energy 179 (2021) 2016–2035, <https://doi.org/10.1016/j.renene.2021.08.024>.
- [9] C. Cerezo Davila, C.F. Reinhart, J.L. Bemis, Modeling Boston: a workflow for the efficient generation and maintenance of urban building energy models from existing geospatial datasets, Energy 117 (2016) 237–250.
- [10] L.G. Swan, V.I. Ugursal, Modeling of end-use energy consumption in the residential sector: a review of modeling techniques, Renew. Sustain. Energy Rev. 13 (2009) 1819–1835, <https://doi.org/10.1016/j.rser.2008.09.033>.
- [11] Y. Vogt, Top-down energy modeling, Strat. Plan. Energy Environ. 22 (2003) 64–79, <https://doi.org/10.1080/10485230309509626>.
- [12] M. Kavgić, A. Mavrogiani, D. Mumovic, A. Summerfield, Z. Stevanovic, M. Djurovic-Petrovic, A review of bottom-up building stock models for energy consumption in the residential sector, Build. Environ. 45 (2010) 1683–1697, <https://doi.org/10.1016/j.buildenv.2010.01.021>.
- [13] C.F. Reinhart, C. Cerezo Davila, Urban building energy modeling - a review of a nascent field, Build. Environ. 97 (2016) 196–202, <https://doi.org/10.1016/j.buildenv.2015.12.001>.
- [14] U. Ali, M.H. Shamsi, C. Hoare, E. Mangina, J. O'Donnell, Review of urban building energy modeling (UBEM) approaches, methods and tools using qualitative and quantitative analysis, Energy Build. 246 (2021) 111073, <https://doi.org/10.1016/j.enbuild.2021.111073>.
- [15] O.F. Ayilara, L. Zhang, T.T. Sajobi, R. Sawatzky, E. Bohm, L.M. Lix, Impact of missing data on bias and precision when estimating change in patient-reported outcomes from a clinical registry, Health Qual. Life Outcomes 17 (2019) 1–9, <https://doi.org/10.1186/s12955-019-1181-2>.
- [16] R. Mohammadizazi, S. Copeland, M.M. Bilec, Urban building energy model: database development, validation, and application for commercial building stock, Energy Build. 248 (2021) 111175, <https://doi.org/10.1016/j.enbuild.2021.111175>.
- [17] Z. Liu, X. Zhou, W. Tian, X. Liu, D. Yan, Impacts of uncertainty in building envelope thermal transmittance on heating/cooling demand in the urban context, Energy Build. 273 (2022) 112363, <https://doi.org/10.1016/j.enbuild.2022.112363>.
- [18] Y.Q. Ang, Z.M. Berzolla, C.F. Reinhart, From concept to application: a review of use cases in urban building energy modeling, Appl. Energy 279 (2020) 115738, <https://doi.org/10.1016/j.apenergy.2020.115738>.
- [19] N. Asim, M. Badiei, M. Mohammad, H. Razali, A. Rajabi, C.H. Lim, M. Ghazali, Sustainability of heating, ventilation and air-conditioning (HVAC) systems in buildings—an overview, Int. J. Environ. Res. Public Health 19 (2022) 1016, <https://doi.org/10.3390/ijerph19021016>.
- [20] D. D'Agostino, F. Minichiello, F. Petito, C. Renno, A. Valentino, Retrofit strategies to obtain a NZEB using low enthalpy geothermal energy systems, Energy 239 (2022) 122307, <https://doi.org/10.1016/j.energy.2021.122307>.
- [21] J. Page, D. Robinson, N. Morel, J.L. Scartezzini, A generalised stochastic model for the simulation of occupant presence, Energy Build. 40 (2008) 83–98, <https://doi.org/10.1016/j.enbuild.2007.01.018>.
- [22] D. Yan, W. O'Brien, T. Hong, X. Feng, H. Burak Gunay, F. Tahmasebi, A. Mahdavi, Occupant behavior modeling for building performance simulation: current state and future challenges, Energy Build. 107 (2015) 264–278, <https://doi.org/10.1016/j.enbuild.2015.08.032>.
- [23] S. Dabirian, K. Panchabikesan, U. Eicker, Occupant-centric modeling for urban building energy simulation: approaches inputs, and data sources - a review, Energy Build. 257 (2021) 111809, <https://doi.org/10.1016/j.enbuild.2021.111809>.
- [24] I.G. Dino, E. Kalfaoglu, O.K. Iseri, B. Erdogan, S. Kalkan, A.A. Alatan, Vision-based estimation of the number of occupants using video cameras, Adv. Eng. Inform. 53 (2022) 101662, <https://doi.org/10.1016/j.aei.2022.101662>.
- [25] M. Mosteiro-Romero, I. Hischier, J.A. Fonseca, A. Schlueter, A novel population-based occupancy modeling approach for district-scale simulations compared to standard-based methods, Build. Environ. 181 (2020) 107084, <https://doi.org/10.1016/j.buildenv.2020.107084>.
- [26] M. Martin, N.H. Wong, D.J.C. Hii, M. Ignatius, Comparison between simplified and detailed EnergyPlus models coupled with an urban canopy model, Energy Build. 157 (2017) 116–125, <https://doi.org/10.1016/j.enbuild.2017.01.078>.
- [27] ASHRAE, ASHRAE Standard 90.1-2013 – Energy Standard For Buildings Except Low-rise Residential Buildings, 2013.
- [28] Y. Chen, T. Hong, Impacts of building geometry modeling methods on the simulation results of urban building energy models, Appl. Energy 215 (2018) 717–735, <https://doi.org/10.1016/j.apenergy.2018.02.073>.
- [29] W. Tian, J. Song, Z. Li, P. De Wilde, Bootstrap techniques for sensitivity analysis and model selection in building thermal performance analysis, Appl. Energy 135 (2014) 320–328, <https://doi.org/10.1016/j.apenergy.2014.08.110>.
- [30] J.A. Gärtner, F.M. Gray, T. Auer, Assessment of the impact of HVAC system configuration and control zoning on thermal comfort and energy efficiency in flexible office spaces, Energy Build. 212 (2020) 109785, <https://doi.org/10.1016/J.ENBUILD.2020.109785>.
- [31] C. Wang, M. Ferrando, F. Causone, X. Jin, X. Zhou, X. Shi, An innovative method to predict the thermal parameters of construction assemblies for urban building energy models, Build. Environ. 224 (2022) 109541, <https://doi.org/10.1016/j.buildenv.2022.109541>.
- [32] E. Pratavera, J. Vivian, G. Lombardo, A. Zarrella, Evaluation of the impact of input uncertainty on urban building energy simulations using uncertainty and sensitivity analysis, Appl. Energy 311 (2022) 118691, <https://doi.org/10.1016/j.apenergy.2022.118691>.
- [33] W.C. Lin, C.F. Tsai, Missing value imputation: a review and analysis of the literature (2006–2017), Artif. Intell. Rev. 53 (2020) 1487–1509, <https://doi.org/10.1007/s10462-019-09709-4>.
- [34] Y.Q. Ang, Z.M. Berzolla, S. Letellier-Duchesne, V. Jusiega, C. Reinhart, UBEM.io: a web-based framework to rapidly generate urban building energy models for carbon reduction technology pathways, Sustain. Cities Soc. 77 (2022), <https://doi.org/10.1016/j.scs.2021.103534>.
- [35] C. Wang, M. Ferrando, F. Causone, X. Jin, X. Zhou, X. Shi, Data acquisition for urban building energy modeling: a review, Build. Environ. 217 (2022) 109056, <https://doi.org/10.1016/j.buildenv.2022.109056>.
- [36] J. Sokol, C.C. Davila, C.F. Reinhart, Validation of a Bayesian-based method for defining residential archetypes in urban building energy models, Energy Build. 134 (2017) 11–24, <https://doi.org/10.1016/j.enbuild.2016.10.050>.
- [37] C.K. Wang, S. Tindemans, C. Miller, G. Agugiaro, J. Stoter, Bayesian calibration at the urban scale: a case study on a large residential heating demand application in Amsterdam, J. Build. Perform. Simul. 13 (2020) 347–361, <https://doi.org/10.1080/19401493.2020.1729862>.
- [38] G.C. Rodríguez, A.C. Andrés, F.D. Muñoz, J.M.C. López, Y. Zhang, Uncertainties and sensitivity analysis in building energy simulation using macroparameters, Energy Build. 67 (2013) 79–87, <https://doi.org/10.1016/j.enbuild.2013.08.009>.
- [39] H. Lim, Z. (John) Zhai, Estimating unknown parameters of a building stock using a stochastic-deterministic-coupled approach, Energy Build. 255 (2022) 111673, <https://doi.org/10.1016/j.enbuild.2021.111673>.

- [40] I. Ballarini, S.P. Corgnati, V. Corrado, N. Talà, Definition of building typologies for energy investigations on residential sector by Tabula Iee-Project: application to Italian case studies, *Roomvent* (2011) 19–22.
- [41] I. De Jaeger, G. Reynders, C. Callebaut, D. Saelens, A building clustering approach for urban energy simulations, *Energy Build.* 208 (2020) 109671, <https://doi.org/10.1016/j.enbuild.2019.109671>.
- [42] L. Filogamo, G. Peri, G. Rizzo, A. Giaccione, On the classification of large residential buildings stocks by sample typologies for energy planning purposes, *Appl. Energy* 135 (2014) 825–835, <https://doi.org/10.1016/j.apenergy.2014.04.002>.
- [43] C.S. Monteiro, A. Pina, C. Cerezo, C. Reinhart, P. Ferrão, The use of multi-detail building archetypes in urban energy modelling, *Energy Proc.* 111 (2017) 817–825, <https://doi.org/10.1016/j.egypro.2017.03.244>.
- [44] O. Pasichnyi, J. Wallin, O. Kordas, Data-driven building archetypes for urban building energy modelling, *Energy* 181 (2019) 360–377, <https://doi.org/10.1016/j.energy.2019.04.197>.
- [45] O.K. Iseri, İ.G. Dino, Building Archetype Characterization Using K-Means Clustering in Urban Building Energy Models, in: *International Conference on Computer-Aided Architectural Design Futures*, Springer Singapore, Singapore, 2021: pp. 222–236.
- [46] M.H. Kristensen, R. Choudhary, R.H. Pedersen, S. Petersen, Bayesian calibration of residential building clusters using a single geometric building representation, *Build. Simul. Conf. Proc.* 4 (2017) 2251–2260, <https://doi.org/10.26868/25222708.2017.330>.
- [47] C. Cerezo, J. Sokol, C. Reinhart, A. Al-Mumin, Three methods for characterizing building archetypes in urban energy simulation. A case study in Kuwait City, in: *14th International Conference of IBPSA - Building Simulation 2015, BS 2015, Conference Proceedings*, 2015, pp. 2873–2880.
- [48] C. Nägeli, C. Camarasa, M. Jakob, G. Catenazzi, Y. Ostermeyer, Synthetic building stocks as a way to assess the energy demand and greenhouse gas emissions of national building stocks, *Energy Build.* 173 (2018) 443–460, <https://doi.org/10.1016/j.enbuild.2018.05.055>.
- [49] A. Mastrucci, P. Pérez-López, E. Benetto, U. Leopold, I. Blanc, Global sensitivity analysis as a support for the generation of simplified building stock energy models, *Energy Build.* 149 (2017) 368–383, <https://doi.org/10.1016/j.enbuild.2017.05.022>.
- [50] H. Kang, The prevention and handling of the missing data, *Korean J. Anesthesiol.* 64 (2013) 402–406, <https://doi.org/10.4097/kjae.2013.64.5.402>.
- [51] A. Rahman, V. Srikumar, A.D. Smith, Predicting electricity consumption for commercial and residential buildings using deep recurrent neural networks, *Appl. Energy* 212 (2018) 372–385, <https://doi.org/10.1016/j.apenergy.2017.12.051>.
- [52] Z. Ye, K. Cheng, S.C. Hsu, H.H. Wei, C.M. Cheung, Identifying critical building-oriented features in city-block-level building energy consumption: a data-driven machine learning approach, *Appl. Energy* 301 (2021) 117453, <https://doi.org/10.1016/j.apenergy.2021.117453>.
- [53] D. Leiria, H. Johra, A. Marszał-Pomianowska, M.Z. Pomianowski, P.K. Heiselberg, Using data from smart energy meters to gain knowledge about households connected to the district heating network: a Danish case, *Smart Energy* 3 (2021) 100035, <https://doi.org/10.1016/j.segy.2021.100035>.
- [54] P.E. McKnight, K.M. McKnight, S. Sidani, A.J. Figueredo, *Missing data: A gentle introduction*, Guilford Press, New York, NY, US, 2007.
- [55] T.C. Ministry of Environment, Enerji Kimlik Belgesi (EKB), Ankara, n.d. <https://www.enerjikimlikbelgesi.com/#ekb-nedir>.
- [56] T.S. Institute, *Data Portal for Statistics*, TÜİK, Ankara, 2021.
- [57] P. Tuik, *Population by sex and age group, 1950–2020*, Ankara, 2021.
- [58] D.M. Grossman, H.F. Linskens, *Monographs on theoretical and applied genetic*, 1977.
- [59] S.J. Sheather, M.C. Jones, A reliable data-based bandwidth selection method for kernel density estimation, *J. R. Statist. Soc.: Ser. B (Methodological)* 53 (1991) 683–690, <https://doi.org/10.1111/j.2517-6161.1991.tb01857.x>.
- [60] H. Wan, J. Yoon, V. Srikrishnan, B. Daniel, D. Judi, Population downscaling using high-resolution, temporally-rich U.S. property data, *Cartogr. Geogr. Inf. Sci.* 49 (2022) 18–31. doi:10.1080/15230406.2021.1991479.
- [61] TÜİK, *Türkiye Family Structure Survey*, Ankara, 2021.
- [62] K. Anderson, *Design energy simulation for architects: Guide to 3D graphics*, 2014. doi:10.4324/9781315851433.
- [63] P.B. Purup, S. Petersen, Research framework for development of building performance simulation tools for early design stages, *Autom. Constr.* 109 (2020), <https://doi.org/10.1016/j.autcon.2019.102966>.
- [64] S. Yang, W. Tian, E. Cubi, Q. Meng, Y. Liu, L. Wei, Comparison of sensitivity analysis methods in building energy assessment, *Proc. Eng.* 146 (2016) 174–181, <https://doi.org/10.1016/j.proeng.2016.06.369>.
- [65] H. Brohus, P. Heiselberg, A. Simonsen, K.C. Sørensen, Uncertainty of energy consumption assessment of domestic buildings, *IBPSA 2009 - International Building Performance Simulation Association 2009*, 2009, pp. 1022–1029.
- [66] B.D.B., Crawley, C.O. Pedersen, L.K. Lawrie, F.C. Winkelmann, *EnergyPlus: Energy Simulation Program*, ASHRAE (2000) 49–56.
- [67] M.S. Roudsari, M. Pak, Ladybug: A parametric environmental plugin for grasshopper to help designers create an environmentally-conscious design, in: *Proceedings of BS 2013: 13th Conference of the International Building Performance Simulation Association*, 2013, pp. 3128–3135.
- [68] M. Waskom, seaborn: statistical data visualization, *J Open Source Softw* 6 (2021) 3021, <https://doi.org/10.21105/joss.03021>.
- [69] TSE, TS-825-2013, *Binalarda ısı Yalıtım Kuralları*, Ankara, 2013.
- [70] DOE, ASHRAE, *Energy standard for buildings except low-rise residential buildings*, ASHRAE Standard 4723 (2013).
- [71] R. McNeel, others, *Rhinoceros 3D, Version 6.0*, Robert McNeel & Associates, Seattle, WA, 2010.
- [72] ASHRAE, Standard 90.1-2016. *Energy Standard for Buildings Except Low-Rise Residential Buildings*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc., 2016, doi:10.1108/17506200710779521.
- [73] İ.G. Dino, C.M. Akgül, Impact of climate change on the existing residential building stock in Turkey: an analysis on energy use, greenhouse gas emissions and occupant comfort, *Renew. Energy* 141 (2019) 828–846, <https://doi.org/10.1016/j.renene.2019.03.150>.
- [74] S. Petersen, M.H. Kristensen, M.D. Knudsen, Prerequisites for reliable sensitivity analysis of a high fidelity building energy model, *Energy Build.* 183 (2019) 1–16, <https://doi.org/10.1016/j.enbuild.2018.10.035>.
- [75] *Ecoinvent Centre, Ecoinvent Database*, 2012.
- [76] T. Tamer, I. Gursel Dino, C. Meral Akgul, Data-driven, long-term prediction of building performance under climate change: building energy demand and BIPV energy generation analysis across Turkey, *Renew. Sustain. Energy Rev.* 162 (2022) 112396, <https://doi.org/10.1016/j.rser.2022.112396>.
- [77] C. Meral Akgul, I. Gursel Dino, Climate change impact assessment in residential buildings utilizing RCP4.5 and RCP8.5 scenarios (RCP4.5 ve RCP8.5 iklim senaryolarına göre konutlarda iklim değişikliği etki değerlendirmesi), *J. Faculty Eng. Architect. Gazi Univ.* 35 (2020) 1665–1683, <https://doi.org/10.17341/gazimmfd.534970>.
- [78] B. Atılğan, A. Azapagic, Assessing the environmental sustainability of electricity generation in Turkey on a life cycle basis, *Energies (Basel)* 9 (2016), <https://doi.org/10.3390/en9010031>.
- [79] M. Hamdy, S. Carlucci, P.J. Hoes, J.L.M. Hensen, The impact of climate change on the overheating risk in dwellings—a Dutch case study, *Build. Environ.* 122 (2017) 307–323, <https://doi.org/10.1016/j.buildenv.2017.06.031>.
- [80] ASHRAE, *ASHRAE Standard 55-2004 – Thermal Comfort*, 2004. doi:10.1007/s11926-011-0203-9.